PARTICIPATION IN PERIOPERATIVE THEATRE CLINICAL SERVICES

1 1.1 Introduction to the unit of learning

This unit standard specifies competencies required to provide PT clinical services. It involves preparing emergency trolley, participating in patients and procedure verification, wheeling, inducing, reversal and monitoring of the patient to be operated.

2 1.2 Summary of Learning Outcomes

1. Prepare for Perioperation Theatre clinical services
2. Assist in Perioperation Theatre clinical services
3. Evaluate Perioperation Theatre clinical services
4. Wide-up Perioperation Theatre clinical services

1.2.1 Learning Outcome: Prepare for Perioperation Theatre clinical services

1.2.2 Introduction to the learning outcome

This unit describes the competencies required to Prepare for Perioperation Theatre clinical services. This learning outcome involves, theatre Layout, types of surgeries performed in theatre, basic anesthesia, preparation of emergency trolley, use of drugs in the emergency trolley, verification and reception of patients in theatre, wheeling of patients to the PT table, positioning of patient on the PT table, patient induction and reversal materials, prepping and draping the patient Reporting patient's condition, re-arranging the Operation Theatre, documentation of patients' condition, specimen handling ,decontamination, types of wastes in Operation Theatre and respective disposal methods and last offices services in Operation Theatre.

1.2.3 Performance Standard

1.1 Emergency trolley is identified in accordance with standard operating procedures (SOPs)
1.2 Emergency trolley is prepared according to standard operating procedures
1.3 Drug/resuscitation trolley is prepared in accordance with standard operating procedures
1.4 Patient to be operated is identified according to theatre list, patient identification number (IP No.), patient's file and identification band
1.5 Operation table is prepared according to operation theatre procedures
1.6 Induction and reversal materials are prepared in accordance with procedure to be undertaken and WHO standards

3 Information Sheet

Definition of key terms

Aerosol- Dispersion of fine mist, droplets, or particulate matter into air (transitive verb: aerosolize, to become airborne).

Antisepsis -Prevention of sepsis by the exclusion, destruction, or inhibition of growth or multiplication of microorganisms from body tissues and fluids.

Antiseptics- Inorganic chemical compounds that combat sepsis by inhibiting the growth of microorganisms without necessarily killing them.

Barrier -Material used to reduce or inhibit the migration or transmission of microorganisms in the environment.

Cross-contamination -Transmission of microorganisms from patient to patient and from inanimate objects to patients and vice versa.

Decontamination- Cleaning and disinfecting or sterilizing processes carried out to make contaminated items safe to handle.

Disinfection -Chemical or mechanical destruction of most pathogens rendering an object safe to handle. Fomite Inanimate object that may be contaminated with infectious organisms and that serves to transmit disease.

Pathogenic microorganisms -Microorganisms that cause infectious disease. They can invade healthy tissue through some power of their own or can injure tissue by a toxin they produce.

Sterile -Free of living microorganisms, including all spores.

Sterile field- Area around the site of incision into tissue or site of introduction of an instrument into a body orifice that has been prepared for the use of sterile supplies and equipment. This area includes all furniture covered with sterile drapes and all personnel who are properly attired in sterile garb.

Sterile technique -Methods by which contamination with microorganisms is prevented to maintain sterility throughout the surgical procedure.

a) Theatre Layout

Operation theatre is the area in which surgical operating work is carried out. This area is also known as ‘operating department’ or ‘operation suite’.

The design of the surgical suite offers a challenge to the planning team to optimize efficiency by creating realistic traffic and workflow patterns for patients, visitors, personnel, and supplies. The design also should allow for flexibility and future expansion. Architects consult surgeons, perioperative nurses, and surgical services administrative personnel before designating functional space within the surgical suite.

Location The surgical suite is usually located in an area accessible to the critical care surgical patient areas and the supporting service departments, the central service or sterile processing department, the pathology department, and the radiology department. The size of the hospital is a determining factor because locating every desirable unit or department immediately adjacent to the surgical suite is complex.

Perioperative theatre zones based on sterility

Unrestricted Area

Street clothes are permitted. A corridor on the periphery accommodates traffic from outside, including patients. This area is isolated by doors from the main hospital corridor and elevators and from other areas of the surgical suite. It serves as an outside- to-inside access area (i.e., a transition zone). Traffic, although not limited, is monitored at a central location. The control desk is accessible from the unrestricted area.

Semi restricted Area

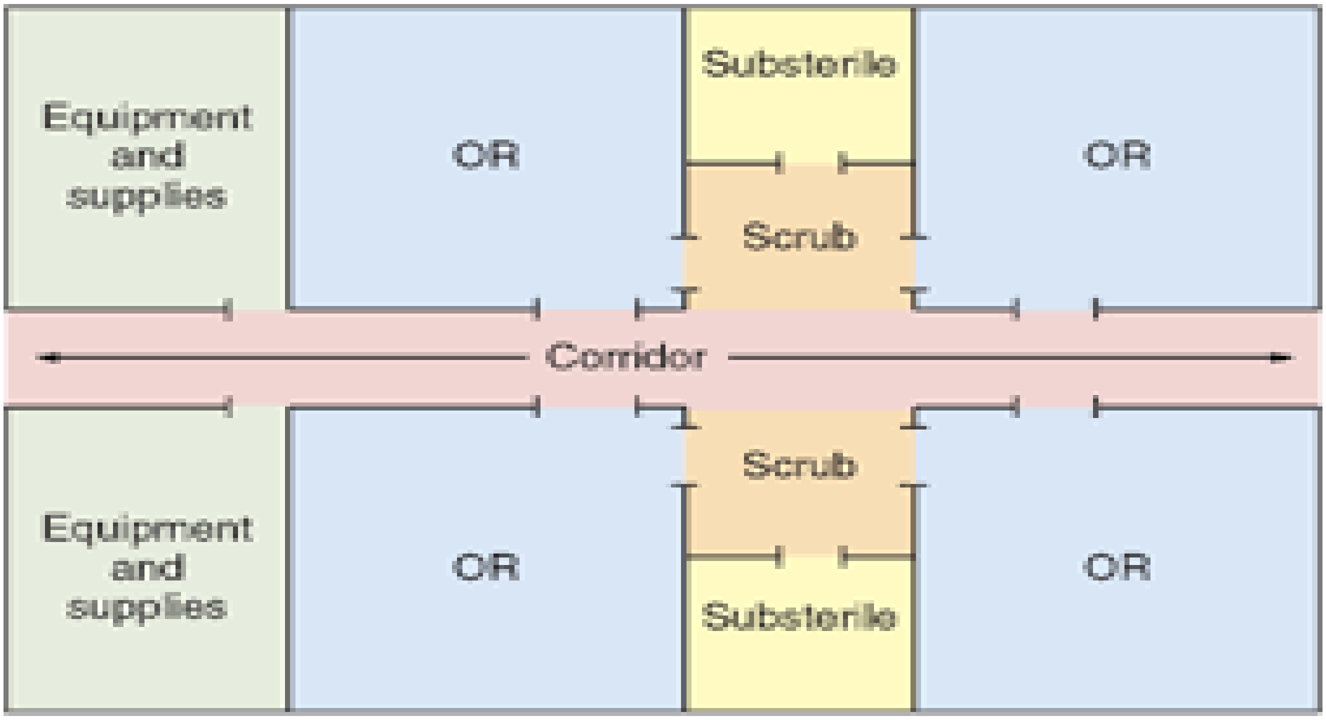
Traffic is limited to properly attired, authorized personnel. Scrub suits and head/beard coverings are required attire. This area includes peripheral support areas, central processing, and access corridors to the operating rooms (ORs). The patient's hair is also covered. Bald heads are covered to prevent distribution of dead skin cells and dander that carry microorganisms.

Restricted Area

Masks are required to supplement OR attire where open sterile supplies and scrubbed personnel are located. Sterile procedures are carried out in the OR and procedure rooms. There are also scrub sink areas and sub sterile rooms or clean core areas where unwrapped supplies are sterilized. Personnel entering this area for short periods, such as laboratory technicians, may wear clean surgical cover gowns or jumpsuits to cover street clothes. Hair/beard covering is worn, and masks are donned as appropriate.

Transition zones both patients and personnel enter the semi restricted and restricted areas of the surgical suite through a transition zone. This transition zone, inside the entrance to the surgical suite, separates the semi restricted OR corridors from the rest of the facility. Masks, caps, shoe covers, and cover gowns (or jumpsuits) may be located on a cart near transition zones adjacent to restricted areas. Nonsurgical personnel who need to enter the restricted zone can don a cover

Figure 1 showing the operation theatre layout



Central corridor, hotel style.

A terminal location is preferred to prevent unrelated traffic from passing through the suite. A location on a top floor is not necessary for microbial control because all air is specially filtered to control dust. Traffic noises may be less evident above the ground floor. Artificial lighting is controllable, so the need for daylight is not a factor; in fact, daylight may be a distraction during the use of video equipment and other procedures that require a darkened environment. Most surgical suites have solid walls without windows. However, some ambulatory surgery rooms have windows to create an ambiance of openness in the suite.

Space Allocation and Traffic Patterns Space is allocated within the surgical suite to provide for the work to be done, with consideration given to the efficiency with which it can be accomplished. The surgical suite should be large enough to allow for correct technique yet small enough to minimize the movement of patients, personnel, and supplies. Provision must be made for traffic control. The type of design predetermines traffic patterns. Everyone—staff, patients,

Functions that can be performed in OT

- Perform surgery in safe, aseptic environment
- Ascertain patient's comforts, both physical & emotional
- Maintain high standard of performance
- Acquire, maintain & suitably utilize equipment
- Maintain theatre discipline by following prescribed procedures & by updating them from time to time
- Attempt maximum utilization of theatre time by proper scheduling
- Prevent iatrogenic complications
- Prevent health hazard from environmental, radiological, anesthetic & infecting agents

- Minimize postponement of surgery

Location of OT

OT is **centralized block** in which all the operating rooms are situated. This arrangement has following advantages:

1. Avoids duplication of expensive equipment
2. Avoids duplication of expensive equipment
3. Avoids need to duplicate routine facilities like changing room, scrub room, surgeon's room etc.
4. Supervision becomes easy
5. It is easier to ensure uniformity & consistency in the procedures
6. Idle theatre time can be reduced

Location of OT should consider following points:

1. Easy approachability, whether it is horizontal traffic or vertical traffic in high rise building.
2. Proximity of central sterile supply department. If that is not possible, it should at least have Proper regional sterile supply section.
3. Proximity to intensive care areas.
4. Proximity to interventional laboratories of super specialty departments

Structure of OT

OT can be divided into clearly demarcated four zones to indicate specific precautions to be practiced before crossing the border of each zone. This zones indicate

- Relatively clean area
- Absolutely clean area
- Absolutely clean & aseptic area
- Unclean area

Outermost Zone: This zone is called as protective zone. This zone is clean, but not sterile area. In this area following activities are housed

- Administrative area
- Operation theatre superintendent or manager
- Office or the anesthesia chief
- Space for surgeons to write or dictate operation notes
- Frozen section biopsy laboratory
- Dark room for developing X-Ray films
- Changing room
- Surgeon's room
- Trolley bay
- Waiting area for relatives

Intermediate zone: This zone is clear, but not sterile. Entry for people bringing supplies, patients etc. can be permitted after changing foot-wear. However, people who have not scrubbed or changed should not be allowed to go to inner zone. It includes following

- Storage area for equipment & instruments
- Supply received from central sterilization & supply department
- Medicines, intravenous fluids, other consumable items and linen
- Post-operative recovery room having 1.5 to 2 beds per operating table
- Preoperative waiting patients

Inner most zone: This area is kept absolutely clean & sterile. No one other than persons actually involved in doing surgery or assisting in surgery should be allowed to enter. In teaching hospitals 4 to 5 students are permitted after changing. This zone includes

- operating room (minimum size 18ft* 20 ft)
- Anesthesia induction room
- Patient holding areas
- Scrub area for surgeons & nurses

Unclean zone for disposal: This zone is used for temporary storage of

- Used linen
- Used instruments
- Waste material
- Cleaning gloves, instruments

Sub areas in OT

Pre-operative check in area (reception): This is important with respect to maintaining privacy, for changing from street clothes to gown and to provide lockers and lavatories for staff.

Staff room: Men and women change dress from street cloth to OT attire.

Holding area: This area is planned for IV line insertion, preparation, catheter / gastric tube insertion, connection of monitors, & shall have O2 and suction lines. Facility for CPR should be available in this area.

Post anesthetic care units (PACU): preferably adjacent to recovery room. These should contain a medication station, hand washing station, nurse station, storage space for stretchers, supplies and monitors / equipment and gas, suction outlets and ventilator. Additionally 80 sq. ft. (7.43 sq. m) for each patient bed, clearance of 5 ft. (1.5 m) between beds and 4 ft. (1.22m) between patient bed sides and adjacent walls should be planned.

Sanitary facility for staff: One wash basin and one western closet (WC) should be provided for 8-10 persons. Showers and their number is a matter of local decision. Inclusion of toilet facilities in changing rooms is not acceptable; they should be located in an adjacent space.

The anesthesia gas / cylinder manifold room / storage area: A definite area to be designated. It should be in a cool, clean room that is constructed of fire resistant materials. Conductive flooring must be present but is not required if non inflammable gases are stored. Adequate ventilation to allow leaking gases to escape, safety labels and separate places for empty and full cylinders to be allocated.

Rest rooms: Pleasant and quiet rest for staff should be arranged either as one large room for all grades of staff or as separate rooms; both have merits. Comfortable chairs, one writing table, a book case etc., may be arranged.

Types of OT complexes

There are three main categories of operating theatre:

1. The single theatre suite with OT, scrub-up and gowning, anesthesia room, trolley preparation, utility and exit bay plus staff change and limited ancillary accommodation.
2. The twin theatre suite with facilities similar to 1, but with duplicated ancillary accommodation immediate to each OT, sometimes sharing a small post anesthesia recovery area.
3. OT complexes of three or more OTs. With ancillary accommodation including post anesthesia recovery, reception, porter's desk, sterile store and staff change.

Ideal OT should have following characteristics

- Operative room for routine work (18ft * 20 ft)
- Super specialty departments like neurosurgery & cardiac surgery require bigger area i.e. about 500ft to 600ft, as these theaters need to have more equipment
- Effecting air conditioning should maintain the temperature as per requirement
- Efficient & sincere paramedical & non-medical staff to carry out necessary instructions promptly.
- Walls: Smooth wall which is impermeable to moisture having finish of epoxy resin or vinyl sheets type of painting.
- Doors: Main door to the OT complex has to be of adequate width (1.2 to 1.5 m). The doors of each OT should be spring loaded flap type, but sliding doors are preferred as no air currents are generated. All fittings in OT should be flush type and made of steel.
- Flooring: The surface / flooring must be slip resistant, strong & impervious with minimum joints (e.g. mosaic with copper plates for antistatic effect) or jointless conductive tiles/ terrazzo, linoleum etc., the recommended minimum conductivity is 1m ohm and maximum 10m Ohms.

Infection Control

- Nurse / surgeon having respiratory infection or finger infection should not participate in surgical procedures.
- It is desirable for surgeons / nurses to wash their hands first with the gloves on. After washing, gloves need to be put in the basin containing sodium hypochlorite solution to prevent risk of spread of infection, particularly HIV. Mask, cap, gowns need to be placed in a plastic drum containing sodium hypochlorite solution.

Types of surgeries in OT

There are various types of surgery and they can be classified based on different parameters.

Based on Timing

Surgery can be classified based on the urgency and need to perform the surgery.

1. **Elective Surgery**

1. The name "elective" might imply that this type of surgery is optional, but that's not always the case. An elective procedure is simply one that is planned in advance, rather than one that's done in an emergency situation. A wide range of surgical procedures can be considered elective surgery. Cosmetic surgeries fall into this category, but so can things like ear tubes, tonsillectomies, and scoliosis surgery. Although these procedures may be done "electively," they can be significant and potentially life-changing operations.

2. **Semi-elective Surgery**

2. Semi-elective surgery is a surgery that must be done to preserve the patient's life, but does not need to be performed immediately.

3. **Urgent Surgery**

An urgent surgery is one that can wait until the patient is medically stable, but should generally be done today or tomorrow

4. **Emergency Surgery**

An emergency surgery is one that must be performed without delay; the patient has no choice other than immediate surgery, if they do not want to risk permanent disability or death.

Based on Body Part

Surgery can also be classified based on the body part that is being operated on.

When surgery is performed on one organ system or structure, it may be classed by the tissue, organ or organ system involved. Examples include:

1. **Gastrointestinal surgery**

This type of surgery is performed within the digestive tract and its accessory organs.

Gastrointestinal surgery involves the treatment for diseases of the parts of the body involved in digestion. This includes the esophagus, stomach, small intestine, large intestine, and rectum. It also includes the liver, gallbladder, and pancreas

2. **Orthopedic surgery**

This is performed on bones or muscles. The medical specialty dedicated to the diagnosis, treatment, rehabilitation, and prevention of injuries of the musculoskeletal system including the muscles, bones, spine, joints, ligaments, tendons, and nerves.

3. **Cardiovascular Surgery**

Cardiovascular surgery is the surgical treatment of diseases and disorders of the heart and blood vessels. Examples of cardiovascular surgery include open heart surgery such as coronary artery bypass grafting, heart valve replacements, heart transplant and vascular surgery such as abdominal aortic aneurysm repairs and femoral-tibial bypass grafting.

4. **Gynaecology**

Gynaecology refers to the surgical specialty dealing with the health of the female reproductive system (uterus, vagina and ovaries).

5. **Neurosurgery**

The surgical specialty involved in the treatment of disorders of the brain, spinal cord, and peripheral nerves.

6. **Ophthalmology (Eyes)**

A medical specialty focusing on the diagnosis and treatment of disorders of the eye. Conditions include cataracts, glaucoma, and conjunctivitis, among others.

7. **Oral and Maxillofacial Surgery**

A dental specialty focusing on the diagnosis and surgical treatment of diseases, injuries and defects of the mouth, teeth and gums. Examples of oral surgery include removal of wisdom teeth, facial injury repair, cleft lip and palate repair, and repair of uneven jaws.

8. **Otolaryngology – Ears, Nose and Throat (ENT)**

Otolaryngology is the medical and surgical treatment of the head and neck, including ears, nose and throat.

9. **Thoracic**

This is also known as cardiothoracic surgery. This is the field of medicine involved in surgical treatment of organs inside the thorax (the chest) generally treatment of conditions of the heart (heart disease) and lungs (lung disease).

10. **Urology**

This is a type of surgery that is performed on the urinary tract. It is a surgical specialty that deals with problems affecting the urinary tract including kidneys, bladder, and prostate.

Based on Equipment Used

Surgery can also be classified based on the type of surgical equipment used for the procedure.

Based on degree of invasiveness

Surgery can also be classified based on the degree of invasiness of the surgical procedure. By invasion, this refers to the medical procedure in which the part of the body is entered by incision.

1. **Minimally Invasive Surgery**

Minimally invasive surgery is any technique involved in surgery that does not require a large incision. This relatively new approach allows the patient to recuperate faster with less pain. Not all conditions are suitable for minimally invasive surgery. Many surgery techniques now fall under minimally invasive surgery: Laparoscopy, Endoscopy, Arthroscopy and many others.

2. **Open Surgery**

An “open” surgery means the cutting of skin and tissues so that the surgeon has a full view of the structures or organs involved. Examples of open surgery are the removal of the organs, such as the gallbladder or kidneys.

Based on purpose

Surgery can also be classified based on the purpose for which the surgery is being performed. Surgeries are performed for various purposes and such can be classified based on the purpose for the surgery.

1. **Exploratory surgery**

Exploratory surgery is a diagnostic method used by doctors when trying to find a diagnosis for an ailment. It is used most commonly to diagnose or locate cancer in humans.

2. **Therapeutic surgery**

Therapeutic surgery treats a previously diagnosed condition

3. **Cosmetic surgery**

Cosmetic surgery is where a person chooses to have an operation, or invasive medical procedure, to change their physical appearance for cosmetic rather than medical reasons.

b) Basic pharmacology

Pharmacology is the study of the interaction of chemicals with living systems.

Drugs are chemicals that act on living systems at the chemical (molecular) level.

Medical pharmacology is the study of drugs used for the diagnosis, prevention, and treatment of disease.

Toxicology is the study of the untoward effects of chemical agents on living systems. It is usually considered an area of pharmacology.

Pharmacodynamic properties of a drug describe the action of the drug on the body, including receptor interactions, dose-response phenomena, and mechanisms of therapeutic and toxic action.

Pharmacokinetic properties describe the action of the body on the drug, including absorption, distribution, metabolism, and excretion. Elimination of a drug may be achieved by metabolism or by excretion.

The nature of drugs

Size. The great majority of drugs lie in the range from molecular weight 100 to 1,000. Drugs in this range are large enough to allow selectivity of action and small enough to allow adequate movement within the various compartments in the body.

Chemistry and reactivity. Drugs may be small, simple molecules (amino acids, simple amines, organic acids, alcohols, esters, ions, etc.), carbohydrates, lipids, or even proteins. Binding of drugs to their receptors.

Absorption of Drugs. Drugs usually enter the body at sites remote from the target tissue and are carried by the circulation to the intended site of action. Before a drug can enter the bloodstream, it must be absorbed from its site of administration. The rate and efficiency of absorption differs depending on the route of administration.

Common routes of administration of drugs and some of their features.

1. **Oral (swallowed).**

Maximum convenience but may be slower and less complete than parenteral (non-oral) routes. Dissolution of solid formulations (e. g tablets) must occur first. The drug must survive exposure to stomach acid. This route of administration is subject to the first pass effect (metabolism of a significant amount of drug in the gut wall and the liver, before it reaches the systemic circulation).

2. **Sublingual (under the tongue).**

Permits direct absorption into the systemic venous circulation thus avoiding the first pass effect. May be fast or slow depending on the physical formulation of the product. Nitroglycerin is administered by this route in the treatment of angina.

3. **Rectal (suppository).**

Same advantage as sublingual route; larger amounts are feasible. Useful for patients who cannot take oral medications (e.g. because of nausea and vomiting).

4. **Intramuscular.**

Absorption is sometimes faster and more complete than after oral administration. Large volumes (e.g. 5 - 10 mL) may be given. Requires an injection. Generally more painful than subcutaneous injection. Vaccines are usually administered by this route.

5. **Subcutaneous.**

Slower absorption than intramuscular. Large volumes are not feasible. Requires an injection. Insulin is administered by this route.

6. **Inhalation**

For respiratory diseases, this route deposits drug close to the target organ; when used for systemic administration (e.g., nicotine in cigarettes, inhaled general anesthetics) it provides rapid absorption because of the large surface area available in the lungs.

7. **Topical**

Application to the skin or mucous membrane of the nose, throat, airway, or vagina for a local effect. It is important to note that topical drug administration can result in significant absorption of drug into the systemic circulation. Drugs used to treat asthma are usually administered this way.

8. **Transdermal**

Application to the skin for systemic effect. Transdermal preparations generally are patches that stick to the skin and are worn for a number of hours or even days. To be effective by the transdermal route, drugs need to be quite lipophilic. Nicotine is available as a transdermal patch for those who are trying to stop cigarette smoking.

9. **Intravenous.**

Instantaneous and complete absorption (by definition, 100%); potentially more dangerous because the systemic circulation is transiently exposed to high drug concentrations.

Distribution of Drugs

i. The distribution of drugs from the site of absorption, through the bloodstream and to the target tissue depends upon:

ii. The blood flow to the tissue is important in the rate of uptake of a drug. Tissues that receive a high degree of blood flow (e.g., brain, kidney) have a fast rate of uptake whereas tissues with a low degree of blood flow (e.g., adipose tissue) accumulate drug more slowly.

iii. Solubility of the drug in the tissue. Some tissues, e.g., brain, have a high lipid content and dissolve a higher concentration of lipophilic agents.

iv. Binding of the drug to macromolecules in the blood or tissue limits their distribution.

v. The ability to cross special barriers. Many drugs are poorly distributed to the brain and the testis because these tissues contain specialized capillaries (the smallest type of blood vessel). The endothelial cells that line these capillaries form a blood-brain barrier and a blood testis barrier by preventing the movement of hydrophilic molecules out of the blood and into the tissue, and by actively pumping lipophilic molecules out of the endothelial cell and into the blood. Of special concern is the ability of drugs to distribute to breast milk in lactating women, and the ability of drugs to cross the placenta (the specialized tissue connecting a pregnant woman and her fetus) and affect the developing fetus. A number of drugs are known to be teratogens (drugs that cause abnormal fetal development) and should be avoided in pregnancy. Women taking drugs that are considered unsafe for infants and that achieve appreciably high concentrations in breast milk should not breast-feed their infants. Information about the safety of drugs in

Elimination of Drugs

The rate of elimination (disappearance of active drug molecules from the bloodstream or body) is almost always related to termination of pharmacodynamics effect. Therefore, knowledge of plasma concentrations of a drug is important in describing the intensity and duration of a drug's effect. There are two major routes of elimination:

1. Excretion. The most common route for drug excretion is through the kidney and out of the body in the urine. To be excreted by the kidney, drugs need to be reasonably hydrophilic so that they will remain in the fluid that becomes the urine. Patients with impaired kidney function usually have a reduced ability to eliminate hydrophilic drugs. To avoid excessively high drug concentrations in these patients, you will need to reduce their dosages or give dosages less frequently. A few drugs enter the bile duct and are excreted in the feces.

Metabolism.

The action of many drugs, especially lipophilic compounds, is terminated by enzymatic conversion, or metabolism, to biologically inactive derivatives. In most cases, the enzymatic conversion forms a more hydrophilic compound that can be more readily excreted in the urine. Most of the enzymes that catalyze drug-metabolizing reactions are located in the gastrointestinal tract and the liver. Some drugs inhibit drug-metabolizing enzymes and thus cause drug-drug interactions when co-administered with drugs that depend upon metabolism for elimination

Classification of drugs

Antibiotics

Are agents/drugs used to treat diseases caused by microorganisms-bacteria, viruses, fungi, protozoa.

Overuse and inappropriate use of antibiotics has fueled a major increase in prevalence of multidrug-resistant pathogens

Antimicrobial Prophylaxis

Should be used in cases where efficacy has been demonstrated and benefits outweigh the risks of prophylaxis.

Surgical Prophylaxis

Surgical wound infections are a major category of nosocomial infections

General principles of antimicrobial surgical prophylaxis:

a) The antibiotic should be active against common surgical wound pathogens; unnecessarily broad coverage should be avoided.
b) The antibiotic should have proved efficacy in clinical trials.
c) The antibiotic must achieve concentrations greater than the minimum inhibitory concentration (MIC) of suspected pathogens, and these concentrations must be present at the time of incision.

Analgesics

- Norciception involves periphral and central mechanisms including an emotional component
- Mediators of pain include: serotonin, prostaglandins
- Analgesics inhibit, mimic or potentiate natural mediators of pain
- Pain receptors on the skin are naked nerve endings
- Sites of actions of analgesics include:
  - Site of injury e.g. NSAIDS
  - Periphral nerves e.g local anaesthetics
  - Thalamus – by closing ‘’gates’’ of the thalamus e.g. opiods, TCAs
  - Some analgesics alter the central appreciation of pain e.g opiods,

Classes of analgesics:

- NSAIDS
- Opiod analgesics
- Anaesthetic agents
  - Local
  - General
- Other drugs with some analgesic effect
  - TCAs

Non-steroidal anti-inflammatory drugs

- These drugs also exert antipyretic and analgesic effects.
- Their anti-inflammatory properties makes them most useful in the management of disorders in which pain is related to the intensity of the inflammatory process.
- NSAIDs are grouped in several chemical classes:
  - Propionic acid derivatives e.g. ibuprofen
  - Indole derivatives e.g. indomethacin
  - Phenylacetic acid derivatives e.g. diclofenac
  - Fenamates e.g meclofenamic acid, mefenamic acid
  - Oxicams e.g. piroxicam
- Most of NSAIDS are well absorbed, and food does not substantially change their bioavailability.
- Most of the NSAIDs are highly metabolized in the liver by Cytochrome P450 enzymes
- Most are excreted renally while some undergo enterohepatic circulation
- NSAIDS are highly protein-bound (upto 98% for some)

Mechanism of action of NSAIDS

The anti-inflammatory/analgesic activity of the NSAIDs is mediated chiefly through inhibition of biosynthesis of prostaglandins

This is achieved through selective or non-selective inhibition of cyclooxigenase (COX) 1 and 2, the enzymes involved in the biosynthesis of prostaglandins

Opioid analgesics

Morphine, the prototypical opioid agonist, has long been known to relieve severe pain with remarkable efficacy

Endogenous opiods include:

- Endorphins
- Enkephalins
- Dynorphins

MOA- Opioids bind to opioid receptors (i.e. μ, δ and κ) located in brain and spinal cord regions involved in the transmission and modulation of pain.

- Classification of opioids
  i. Strong agonists
- Morphine, hydromorphone, and oxymorphone are strong agonists useful in treating severe pain
- Pethidine- Does not inhibit uterine contraction thus can be used to control obstetric pain. Can cause respiratory depression in neonates
- Heroin (diamorphine, diacetylmorphine) is potent and fast-acting, but its use is prohibited
- Methadone
- Fentanyl, sufentanil, alfentanil, and remifentanil

d) Basic anaesthesia

Anaesthesia means 'loss of sensation'

It is defined as a temporary state consisting of unconsciousness, loss of memory, lack of pain and muscle relaxation

Anaesthetic agents are classified into two:

i. Local anaesthetics
ii. General anaesthetics

Local anaesthetics

Cocaine, procaine, lidocaine/lignocaine, bupivacaine, benzocaine

MOA - reversibly block impulse conduction along nerve axons and other excitable membranes that utilize sodium channels as the primary means of action potential generation.

Clinically, local anesthetics are used to block pain sensation from (or sympathetic vasoconstrictor impulses to) specific areas of the body

- Systemic absorption of injected local anesthetic from the site of administration is determined by
  - Dosage
  - Site of injection
  - Drug-tissue binding
  - Local blood flow
  - Use of vasoconstrictors (eg, epinephrine), and
  - The physicochemical properties of the drug itself

Local anesthetics are converted in the liver or in plasma to more water-soluble metabolites and then excreted in the urine

Toxicity

- Two major forms of local anesthetic toxicity are:

Systemic effects following absorption of the local anesthetic from their site of administration and

Direct neurotoxicity from the local effects of these drugs when administered in close proximity to the spinal cord and other major nerve trunks

Lignocaine

Is the most widely used LA

Has a quick onset and medium duration of action

Can be administered topically as a jell or aerosol

Is used in all forms of local anesthesia

Bupivacaine

Is a long acting LA commonly used for epidural and spinal anaesthesia

Has a slow onset of action with a duration of action between 5-12hrs

Epidural half-life is much shorter (2hrs) but still longer than lignocaine

It is the LA of choice in obstetric surgeries

May have acute CNS toxicity like lignocaine

Rights of drug administration

Safe drug and pharmaceutical administration in the OR is practiced according to the seven rights of medication administration in any setting:

- Right patient
- Right drug
- Right timing (cements/glues)(antibiotics)
- Right dose
- Right route
- Right reason
- Right documentation

e) Emergency trolley and its purpose

Crash cart or code cart is a set of trays/drawers/shelves on wheels used in hospitals for transportation and dispensing of emergency medication/equipment at site of medical/surgical emergency for life support protocols to potentially save someone's life. The cart carries instruments for cardiopulmonary resuscitation and other medical supplies while also functioning as a support litter for the patient.

The contents of a crash cart vary from hospital to hospital, but typically contain the tools and drugs needed to treat a person in or near cardiac arrest. These include but are not limited to:

- Monitor/defibrillators, suction devices, and bag valve masks (BVMs) of different sizes
- Advanced cardiac life support (ACLS) drugs such as epinephrine, atropine, amiodarone, lidocaine, sodium bicarbonate, dopamine, and vasopressin
- First line drugs for treatment of common problems such as: adenosine, dextrose, epinephrine for IM use, naloxone, nitroglycerin, and others
- Drugs for rapid sequence intubation: succinylcholine or another paralytic, and a sedative such as etomidate or midazolam; endotracheal tubes and other intubating equipment
- Drugs for peripheral and central venous access
- Pediatric equipment (common pediatric drugs, intubation equipment, etc.)
- Other drugs and equipment as chosen by the facility

Equipment

- Airway (oral and nasal) all sizes
- McGill forceps, large and small
- King Airway set (3) eliminates the need for laryngoscope and endotracheal tubes
- Bag valve mask (adult and pediatric)
- Nasal cannula
- Non rebreather oxygen face masks (3 sizes)
- IV start packs
- Normal saline solution (1000ml bags)
- IV tubing
- Angiocaths (various sizes)
- 10ml normal saline flush syringes (3)
- Gauze
- Alcohol preps
- Monitor with defibrillator (preferred) or AED
- Syringe nasal adaptor (nasal narcan atomizer)
- A checklist confirming everything that should be on the cart (print this page, or buy our laminated checklist)

Drugs

- Aspirin 81mg Tablets

- Nitroglycerin spray or 0.4mg tablets
- Dextrose 50% (dextrose 25% if treating pediatrics)
- Narcan 1mg/ml (6)
- Epinephrine 1:10,000 Abboject
- Atropine Sulfate 1mg Abboject
- Amiodarone 150mg Vial
- EpiPen
- EpiPen Jr
- Solumedrol 125mg vial
- Benadryl 50mg vial (2)
- Adenosine 6mg (4)
- Lopressor 10mg (2)
- Cardiazem 20mg vial (2)
- Pronestyl (procainamide) 1g in 10 ml 100mg/ml Vial (1)

Why a Crash Cart?

Any environment in which a patient may unexpectedly experience a medical emergency needs to have the equipment to deal with that emergency efficiently. That's the job of a crash cart. A crash cart contains the equipment and medications that would be required to treat a patient in the first thirty minutes or so of a medical emergency. Although crash carts can differ somewhat depending upon their location, the basic crash cart will contain similar equipment.

Figure 1: Diagrams showing a crush cart

(a) (b)

Maintenance of Crash Cart

The worst thing ever is to reach for a piece of emergency equipment or an emergency medication and find it inoperable or expired. It is important that the crash cart be checked regularly and maintained so that its contents are there when needed.

The following is a maintenance routine that should be completed at least monthly:

Expiration dates on medications should be checked on the first day of the month

Expired medications should be promptly removed and replaced

The defibrillation pads on the AED or the defibrillator should be checked for expiration date

The battery charge on the monitor and/or AED should be checked and documented

Use of drugs in the emergency trolley

Allergy:

To treat allergy and hypersensitivity following drugs are commonly used in emergency department of a hospital

1. Hydrocortisone sodium succinate injection, 100mg powder for reconstitution

2. Adrenaline 1 in 1,000 (1mg/mL)
3. Chlorphenamine tablets 4mg
4. Chlorphenamine injection 10mg/mL

To reduce pain (Painkillers):

1. Paracetamol tablets 500mg
2. Dihydrocodeine tablets 30mg
3. Cyclimorph® injection
4. Diclofenac 75mg/3mL

Anti-Narcotic

Naloxone injection 400 micrograms/mL

Antibiotics:

Patients may encounter a variety of infections which may be caused by pathogenic micro-organisms.

- Amoxicillin capsules/co-amoxiclav tablets and suspension
- Cefuroxime injection
- Cefalexin capsules
- Erythromycin tablets and mixture
- Trimethoprim tablets
- Metronidazole tablets
- Penicillin V tablets
- Benzylpenicillin injection 600mg vial

Cardiac Drugs:

- Aspirin tablets 300mg
- Atropine injection 600 micrograms/mL
- Digoxin tablets 250 micrograms
- Furosemide injection 50mg/5mL
- Lidocaine 2%, 5mL (100mg bolus)
- Glyceryl trinitrate spray
- Tenecteplase vial
- Dalteparin injection

Central Nervous System:

- Diazepam injection 10mg/2mL
- Diazepam tablets 5mg
- Diazepam rectal tubes 5mg/2.5mL
- Chlorpromazine injection 50mg/2mL
- Chlorpromazine tablets 25mg
- Haloperidol injection 5mg/mL

Respiratory System:

- Aminophylline injection 250mg/10mL
- Hydrocortisone sodium succinate injection, 100mg powder for reconstitution
- Prednisolone tablets 5mg
- Salbutamol or terbutaline inhaler
- Salbutamol for nebuliser
- Ipratropium for nebuliser

f) Aseptic techniques

It is impossible to exclude all microorganisms from the environment, but for the safety of both patients and personnel, every effort is made to minimize and control these microorganisms. Microorganisms are present in the air and on animate and inanimate objects at all times. To effectively apply the principles of asepsis, environmental control, and sterile techniques discussed in this chapter, the meaning of terms related to aseptic technique must first be understood. Therefore the basis of prevention is the knowledge of causative agents and their control, as well as the principles of aseptic and sterile techniques. The methods by which microbial contamination is contained in the environment are referred to as aseptic technique. The OR in the restricted area is aseptic at best because the room and air cannot ever be 100% free of microbial content. Some key elements of asepsis include the following facts:

- Aseptic technique is sometimes referred to as clean technique.
- Items have been cleaned and decontaminated so they are safe to handle with clean, bare hands.
- Items in use in patient care are handled with examination gloves for the protection of both the caregiver and the patient.
- Items have been cleaned, decontaminated, disinfected, or terminally sterilized without a wrapper, and stored in a clean, dry place.
- Items may start out sterile but are not maintained or used under sterile conditions. Skin preps may be packaged sterile, but skin cannot be sterilized. The process is aseptic.
- Contamination is contained. Extraneous contamination is avoided.
- Items are set up on clean towels or drapes and used with examination gloves.
- Items are not sterile or maintained sterile during use. Extraneous contamination is avoided.
- Disposable items are not cleaned and reused for another patient.
- Items are classified as semi critical or noncritical by Spaulding's classification of the importance of patient care items.
- Items can be used outside the confines of the restricted area.
- Items are used on intact skin or mucous membranes.
- Items are not used when the patient's vascular system will be entered.

Sterile Technique

Sterile technique incorporates many processes associated with asepsis but to a higher, more controlled degree. In sterile technique all microorganisms must be maintained at an irreducible minimum meaning as low as absolutely possible.

Keep in mind that a sealed sterile package must be opened at some point during patient care to use the contents. That means opening the package to the room air. Even in the restricted area the room air has a microbial count that we cannot eliminate. We try as hard as possible to minimize the room traffic and maintain environmental controls to maintain the sterile field. Essential elements of sterile technique include the following factors:

- Items are used only in a sterile field in the restricted area.
- Items are used by sterile team members wearing appropriate sterile attire.
- Items are used in areas of the patient's body where the site has been prepped.
- Items may be used in invasive surgery.
- Items may be used in body areas with non-intact skin and membranes and may enter the patient's vascular system.
- Items are classified as critical according to Spaulding's level of importance of patient care items.
- Items have been cleaned, decontaminated, and packaged before sterilization.
- Items processed by sterilization are stored wrapped and remain so until use by a sterile team member.
- Items are for individual patient use only. Reusable items can be reprocessed and resterilized for use with another patient. Disposable items are discarded after use. If opened and unused, disposable items are discarded.
- Items that become contaminated are discarded and replaced immediately.

Patient Identification

Preadmission Procedures Some preoperative preparations can be performed in the surgeon's office. Patients are then referred to the preoperative testing center of the hospital or ambulatory care facility. Tests and records should be completed and available before the patient is admitted the day of the surgical procedure. Preadmission tests (PATs) are scheduled according to the guidelines of each facility. Some tests are acceptable only for a 30-day period or are repeated before admission. The type of testing performed depends on the patient's known or suspected condition and the complexity of the surgical procedure. The preoperative preparations include:

1. Medical history and physical examination. These are performed and documented by a physician, nurse practitioner, physician assistant (PA), or the registered nurse first assistant (RNFA). Allergies and sensitivities should be noted. The preoperative nurse establishes the baseline for the patient's vital signs
2. Laboratory tests. Testing should be based on specific clinical indicators or risk factors that could affect surgical management or anesthesia. Tests include age, sex, preexisting disease, magnitude of surgical procedure, and type of anesthesia. Ideally

these tests should be completed 24 hours before admission so the results are available for review. Some facilities perform laboratory studies the morning of the procedure.

Blood type and cross match. If a transfusion is anticipated, the patient's blood is typed and cross matched. Many patients prefer to have their own blood drawn and stored for auto transfusion. Patients should be advised that blood banks charge an additional fee to store and preserve autologous blood for personal use. Even if the patient is to have an auto transfusion, his or her blood should still be typed and cross matched in the event that additional transfusions are needed. If the patient refuses to accept blood transfusions, the appropriate documentation of refusal should be completed according to the policies and procedures of the facility

Self-assessment

Written assessment

1. Name five emergency drugs found in the crush cart that can be used on respiratory system diseases
2. What are the rights of drug administration to ensure safe drug administration?
3. Name three classifications of drugs used in pain management/analgesics
4. What is the ideal location of an operation theatre
5. State 5 operation table accessories
6. Name five intravenous anesthesia inducing agents

Practical assessment

a) Conduct a hospital visit in an operation theatre;

1. Examine a crash cart and write down a report; include all the drugs and equipment required for emergency and use of the crash cart in emergency.
2. On the flash cards provided, draw a diagram to illustrate the operation theatre lay out for the Operation visited; identify the important zones and areas
3. Perform a role play in groups of twos demonstrating how patient identification is performed before surgery.

Tools, Equipment, Supplies and Materials

Operating theatre/Skills laboratory

Crush cart

Gloves

Flash cards

Operation theatre table

Scrubs

Surgical head cover

Desktops/computers/laptops

Projectors

Mannequin

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Learning Outcome 2: Assist in Peri-operative Theatre clinical services

Introduction to the learning outcome

This learning outcome involves receiving the patient to the operating room, verifying patient using the safety checklist, wheeling the patient to operating room, positioning the patient to the operation table, handing over inducing agents to anesthetist, monitoring the patient and assisting in patient reversal.

1.2.3 Performance Standard

1.1 Patient is verified according to procedure to be undertaken, theatre list, patient identification number (IP No.), patient's file and identification band

1.2 Patient is received according to Standard Operation Procedures

1.3 Patient to be operated is wheeled to operation table according to Standard Operation Procedures

1.4 Patient is positioned according to surgical procedure to be undertaken

1.5 Induction materials are handed over to the anesthetist according to Standard Operation Procedures and type of anesthesia

1.6 Patient is monitored according to standard operating procedures and doctors' instructions

1.7 Reversal materials are handed over to anesthetist in accordance to type of anesthesia given and Standard Operation Procedures

1.8 Patient is wheeled to Post Anesthesia Care Unit (PACU)/recovery room according to Standard Operating Procedure

1.2.4 Information Sheet

Definition of key terms

PACU - Post Anesthesia Care Unit

Content/procedures/methods/illustrations

Verification and reception of patients in theatre

Admission to the Pre surgical Holding Area

The holding area nurse greets the patient by name and introduces himself or herself. The nurse stands next to the midsection of the stretcher so the patient can comfortably see him or her. The holding area nurse performs the following steps and documents them on the surgical checklist: Places a warm blanket on the patient, verifies patient identification by name and date of birth, and notifies the individual at the surgery control desk that the patient is in the holding area.

Verifies the surgical procedure, site, and surgeon verbally with the patient and/or family as appropriate. The surgical site is validated by observing the indelible ink of the surgeon's initials. Hospital policy is followed for identification of surgical sites that cannot be marked.

i. Reviews the patient's surgical safety checklist completeness.
ii. Medical history and physical examination
iii. Laboratory reports
iv. Consent forms and documentation of consents
v. Informed consent data

vi. General consent to treat
vii. Anesthesia consent
viii. Measures the vital signs and blood pressure.
ix. Verifies allergies and medication history.
x. Checks skin tone and integrity.
xi. Verifies physical limitations.
xii. Notes the patient's mental state.
xiii. Puts a cap on the patient to protect his or her hair, for purposes of asepsis, and to help minimize hypothermia. Patients who are bald are required to wear a head covering to prevent heat loss and scalp dander shed.
xiv. Notifies the individual at the surgery control desk when the patient is ready for transport to the OR. The patient is under constant observation by the patient care staff until transported from the surgical department to another patient care unit or discharged. The holding area nurse records pertinent findings in the perioperative patient care record and on the surgical checklist. If a perioperative patient assessment has not been performed, the holding area nurse will assess the patient, formulate the nursing diagnoses and expected outcomes, and prepare an individualized plan of care.

Preanesthesia Preparations

The anesthesia provider or surgeon may wish to talk with the patient before sedation is given. The patient should be informed about all procedures before they are initiated. The following list of procedures may need to be completed before the induction of anesthesia; some or all of these procedures can be performed in the preoperative holding area if there are adequate facilities for patient privacy:

i. Removal of body hair, if ordered.
ii. Marking of surgical incision sites such as for plastic surgery of the face and torso.
iii. Before the Induction of Anesthesia, the anesthesia provider also has immediate preanesthesia duties, such as:

- Checking and assembling equipment before the patient enters the room. Airways, endotracheal tubes, laryngoscopes, suction catheters, labeled prefilled medication syringes, and other items are arranged on a cart or table.
- Reviewing the preoperative physical examination, history, and laboratory reports in the chart.
- Making certain that the patient is comfortable and secure on the operating bed.
- Checking for denture removal or any loose teeth. The latter may be secured with a 2-0 silk tie and taped to the patient's cheek to prevent possible aspiration.
- Checking to be certain that contact lenses have been removed.
- Asking the patient when he or she last took anything by mouth, including medications.
- Checking the patient's pulse, respiration, and blood pressure to obtain a baseline for the subsequent assessment of vital signs while the patient is anesthetized.

- Listening to the heart and lungs, and then connecting ECG monitor leads and attaching the pulse oximeter and other monitoring devices
- Starting the IV access. This may be done in the holding area or an induction room. Some patients arrive with an IV line in place.
- Preparing for and explaining the induction procedure to the patient. If properly premedicated, the patient should be able to respond to simple instructions.

Wheeling of patients to the PT table

Transportation to the Operating Room Suite Patients may be taken to the OR suite approximately 30 to 45 minutes before the scheduled time of the procedure. For safety they are commonly transported via a transport stretcher or wheelchair. If a stretcher is used, it should be pushed from the head end so the patient's feet go first. Rapid movements through corridors and around corners may cause dizziness and nausea, especially if the patient has been medicated. The attendant at the head end can observe for vomiting or respiratory distress. The patient may be more comfortable if the head end of the stretcher is raised. If transporting by wheelchair, the patient should be instructed not to help with door opening and to keep hands on the lap. A blanket or sheet should be placed on the seat of the chair and a cover over the lap for modesty. It is inappropriate for bare buttocks to be seated on the uncovered surface of the wheelchair. Take care with tubing such as IVs or catheters so they do not get tangled in the wheels. Urinary drainage bags should be maintained below the level of the bladder to prevent reflux infection. Some ambulatory patients may be permitted to walk to the OR. They are given foot protection such as slippers to prevent injury. The slippers should be skid-proof on the bottom. Paper slippers may be unsafe. They do not protect from injury or slippage. The slippers may need to be removed for the procedure but should be retained until the end of the procedure and returned to the patient for use within the facility. Ideally certain elevators are designated "For OR Use Only," which ensures privacy and minimizes microbial contamination.

Transfer to the Operating Room When everything is ready the circulating nurse comes to the holding area for the patient. A hand-off report is given by the holding room nurse to the circulating nurse. The surgical checklist is one way to assure that an accurate exchange of information is given as patient care changes hands.

The circulating nurse will in turn continue documenting on the checklist for continuity and relay the information to the PACU nurse when the surgical procedure is completed and patient care is transferred for post anesthesia recovery. It is advantageous if the circulator is the perioperative nurse who made the preoperative visit; the patient will appreciate seeing a familiar face. Before transporting the patient into the OR, the circulating nurse has several important duties to fulfill:

1. Greet the patient, and validate his or her identity

The circulating nurse should introduce himself or herself if he or she has not previously met the patient. The patient should be addressed as Mr., Mrs., or Ms.—not by the first name. It is appropriate to ask the patient to state and/or

Spell his or her name. The patient should be asked his or her full name and date of birth. When the patient arrives at the facility, an identifying wristband is put on in the admitting office. The holding room nurse checks the band before the patient leaves for the OR. The circulating nurse compares the information on the wristband, including the identification number, with the information on the medical record and with the information on the surgical schedule: name, anticipated surgical procedure, time, and surgeon. An allergy band and other notification band (e.g. Do Not Resuscitate [DNR]) should be checked at the same time.

Identification on the stretcher or bed ensures the patient's return to the same stretcher or bed after surgery, if this is the procedure. If the patient is an infant or child, the identification bracelet may be on the ankle. Any tag on the crib should be out of his or her reach. Always validate the identity of a child with a parent or legal guardian. Ask what procedure is being performed. Validate the surgical site marking as appropriate. Verification of the surgical procedure, site, and surgeon with the patient provides reassurance that this is the correct patient. The patient's own words should be documented on the chart. The circulating nurse should note the presence of the surgeon's identifying mark on the surgical site. If the patient is heavily sedated, the surgeon may be asked to help identify the patient.

2. Check the side rails, restraining straps, IV infusions, and indwelling catheters.
3. Observe the patient for any reaction to the medication.
4. Observe the patient's anxiety level.
5. Check the history and physical data, laboratory tests, x-ray reports, and consent form(s) or documentation in the patient's medical record.
6. Review the plan of care and the surgical checklist. Pay particular attention to allergies and any previous unfavorable reactions to anesthesia or blood transfusions. b. Become familiar with this patient's unique and individual needs. The patient is taken into the OR after the surgeon sees him or her and the anesthesia provider is ready to receive the patient. The main preparations for the procedure should be complete so the circulating nurse can devote undivided attention to the patient.

Positioning of patient on the PT table

Equipment for Positioning OR Bed Many different OR beds with suitable attachments are available, and practice is necessary to master the adjustments. OR beds are versatile and adaptable to a number of diversified positions for many surgical specialties; orthopedic, urologic, and fluoroscopic tables are often used for specialized procedures.

Figure 2: Diagram showing an operating table

Figure 3: Diagram showing an operating table with special attachments

Special Equipment and Bed Attachments
The equipment used in positioning is designed to stabilize the patient in the desired position and thus permit optimal exposure of the surgical site. All devices are clean, free of sharp edges, and padded to prevent trauma or abrasion. Each OR bed has attachments for specific purposes. Many positioning devices to protect pressure points and joints are commercially available. If the devices are reusable, they are washable; some may be terminally sterilized for asepsis between uses.

Safety Belt (Thigh Strap). For restraint of leg movement during surgical procedures, a sturdy, wide strap of durable material (e.g., nylon webbing, conductive rubber) is placed and fastened over the thighs, above the knees, and around the surface of the OR bed.
Shoulder Bridge (Thyroid Elevator). When a shoulder bridge is used, the head section is temporarily removed and a metal bar is slipped under the mattress between the head and body sections of the OR bed. The bridge can be raised to hyperextend the shoulder or thyroid area for surgical accessibility. This position can be achieved also by placing a towel or blanket roll transversely under the shoulders, which causes the neck to hyperextend. A perpendicular towel roll can be placed between the shoulders to cause the shoulders to fall back bilaterally, which elevates the sternum.
Shoulder Braces or Supports. Adjustable well-padded concave metal supports are occasionally used to prevent the patient from slipping when the head of the OR bed is tilted down, such as in the Trendelenburg's position
Body Rests and Braces. Body rests and braces are made of metal and have a foam-rubber or gel pad covered with conductive waterproof fabric. These devices are placed in metal clamps on the side of the OR bed and are slipped in from the edge of the OR bed against the body at various points to stabilize it in a lateral position.
Lateral Positioner (Kidney Rests). Kidney rests are concave metal pieces with grooved notches at the base; they are placed under the mattress on the body elevator flexion of the OR bed. They are slipped in from the edge of the OR bed and placed snugly against the body for lateral stability in the side-lying kidney position. Although the kidney rest is padded, care should be taken so that the upper edge of the rest does not press too tightly against the body.
Body (Hip) Restraint Strap. With a body restraint strap, a wide belt with a padded center portion (to protect the skin) is placed over the patient's hips and secured with hooks to the sides of the OR bed. This strap helps to hold the patient securely in the lateral position. Some surgeons prefer to use 2-inch to 3-inch-wide bands of adhesive tape to secure the shoulders and hips of patients in the lateral position. A towel can be placed over the patient's skin before the tape is applied.
Adjustable Arched Spinal Frame. An adjustable arched spinal frame consists of two padded arches mounted on a frame that is attached to the OR bed. The patient is placed in

the prone position with the ventral surface of the abdomen over this device (referred to as a Wilson frame). The pads extend from the shoulders to the thighs, with the abdomen hanging dependently between the arches. The desired degree of flexion for spinal procedures is achieved by adjusting the height of the arch by means of a crank. Other types of prone positioning frames are available

Stirrups. Metal stirrup posts are placed in holders, one on each side rail of the OR bed, to support the legs and feet in the lithotomy position. The feet are supported with canvas or fabric loops that suspend the legs at a right angle to the feet. These stirrups are sometimes called candy cane or sling stirrups

Metal Footboard. The footboard can be used flat as a horizontal extension of the OR bed or raised perpendicular to the OR bed to support the feet, with the soles resting securely against it. It is padded when the patient is placed in reverse Trendelenburg's position. The patient's full body weight should not rest on the soles of the feet.

Lift Sheet (Draw sheet). A double-layer sheet is placed horizontally across the top of a clean sheet on the OR bed. After patients are transferred to the OR bed, their arms are enclosed in the lower flaps of this sheet, with the palms against the sides in a natural position and the fingers extended along the length of the body

Anterior chest, epigastrium, umbilicus, pelvis Laparoscopy

Prepping and draping the patient

Draping is the procedure of covering the patient and surrounding areas with a sterile barrier to create and maintain an adequate sterile field. An effective barrier eliminates or minimizes passage of microorganisms between nonsterile and sterile areas.

Criteria to be met in establishing an effective barrier are that the material must be:

Blood and fluid resistant to keep drapes dry and prevent migration of microorganisms. Material should be impermeable to moist microbial penetration (i.e., resistant to strike-through) and resistant to tearing, puncture, or abrasion that causes fiber breakdown and thus permits microbial penetration.

Lint-free to reduce airborne contaminants and shedding into the surgical site. Cellulose and cotton fibers can cause granulomatous peritonitis or embolize arteries.

Antistatic to eliminate risk of a spark from static electricity. Material must meet standards of the National Fire Protection Association.

Sufficiently porous to eliminate heat buildup so as to maintain an isothermic environment appropriate for the patient's body temperature.

Drapable to fit around contours of the patient, furniture, and equipment.

Dull and non-glaring to minimize color distortion from reflected light.

Free of toxic ingredients, such as laundry residues and nonfast dyes.

Flame resistant to self-extinguish rapidly on removal of an ignition source. This is a concern with use of lasers, ESUs, and other high-energy devices that provide an ignition source at the sterile field. Drapes become fuel for a fire. Some materials are more flammable than others; some are fire-retardant. Draping Materials Self-Adhering Sheeting. Sterile, waterproof, antistatic, and transparent or translucent plastic sheeting may be applied to dry skin. Incise Drape. The entire clear plastic drape has an adhesive backing that is applied to skin. This may be applied separately, or the sheeting may be incorporated into the drape sheet. The skin incision is made through the plastic. Antimicrobial incise drapes have an antimicrobial agent impregnated in the adhesive or the polymeric film. A film coated with an iodophor-containing adhesive, for example, slowly releases active iodine during the surgical procedure to effectively inhibit proliferation of organisms from the patient's skin. The antimicrobial may be triclosan or another agent that does not contain iodine. The skin may be prepped with alcohol and allowed to dry before the drape is applied. Time is a factor in microbial accumulation from resident bacteria. Antimicrobial incise drapes are used to sustain suppression, particularly for procedures that last more than 3 hours.

Towel Drape. A nonwoven towel drape has a band of adhesive along one edge. Used as a draping towel, it remains fixed on skin without towel clips. This is advantageous when clips might obscure the view of a part exposed to x-rays during the surgical procedure. It also is used to wall off a contaminated area, such as a stoma, from the clean skin area to prevent spilling contents and causing infection or chemical irritation. Aperture Drape. Adhesive surrounds a fenestration (opening) in the plastic sheeting. This secures the drape to the skin around the surgical site, such as an eye or ear. Caution is used in applying this type of drape around the face of a patient who is awake. The patient must have breathing space. If oxygen is used during a local procedure, it can build up under the drape creating a fire hazard if cautery is used. Some patients experience claustrophobia. Fabric towels may not feel as confining. Advantages of Self-Adhering Drape Material

- Resident microbial flora from skin pores, sebaceous glands, and hair follicles cannot migrate laterally to the incision.
- Microorganisms do not penetrate the impermeable material.
- Landmarks and skin tones are visible through the transparent plastic.
- Inert adhesive holds drapes securely, eliminating the need for towel clips and possible puncture of the patient's skin.
- Plastic sheeting conforms to body contours and has elasticity to stretch without breaking its adhesion to skin. Some self-adhering drapes have sufficient moisture-vapor permeability to reduce excessive moisture buildup that could macerate the skin or loosen adhesive. A nonporous material should not cover more than 10% of the body surface because it may interfere with the patient's thermal regulatory mechanism of perspiration evaporation. The heat-retaining property of plastic causes the patient to perspire excessively, but its nonporous nature prevents evaporation. This material is used in the following manner:

1. The usual skin preparation is done.

2. The scrubbed area must be dry. It may dry via evaporation, or excess solution may be blotted with a towel.
3. Transparent plastic material is applied firmly to the skin, with the initial contact along the proposed line of incision. The drape is smoothed away from the incision area.
4. Regular fabric drapes are applied over the plastic sheeting unless plastic is incorporated into the fenestrated area of the drape. Nonwoven Disposable Drapes. Most nonwoven disposable materials are compressed layers of synthetic fibers (i.e., rayon, nylon, or polyester) combined with cellulose (wood pulp) and bonded together chemically or mechanically without knitting, tufting, or weaving. This material may be either nonabsorbent or absorbent. Polypropylene and foil also are used in some types. Laser-Resistant Drape. Nonwoven drapes that contain cellulose ignite and burn easily. Polypropylene drapes do not ignite, but they can melt. An aluminum-coated drape may be safest for use with lasers, especially around the oxygen-enriched environment of the head and neck area.

Thermal Drape. An aluminum-coated plastic body cover reflects radiant body heat back to the patient to reduce intraoperative heat loss. A sterile, nonconductive, radiolucent thermal drape may be used as the final drape. The patient may be wrapped in nonsterile reflective covers (thermodrape) in the preoperative holding area. These may remain in place through the surgical procedure under standard sterile drapes and during postoperative recovery in the post anesthesia care unit (PACU

Lap sheets with a vertical fenestration can be used in some spinal procedures. Specialty lap sheets for cesarean sections may have an oval fenestration. Thyroid Sheet. The thyroid sheet is the same size as a laparotomy sheet. The fenestration is transverse or diamond shaped and is positioned closer to the top of the patient over the neck area.

Chest Sheet. The chest sheet is similar to a laparotomy sheet except that the fenestration provides for a larger exposure. It is used for chest and breast procedures. Some chest sheets have a horizontal fenestration.

Hip Sheet. The hip sheet is similar to the laparotomy sheet but somewhat longer to completely cover the orthopedic fracture table. The fenestration may be oval and used in conjunction with an impervious stockinette. Perineal Sheet. The perineal sheet is of adequate size to create a sterile field with the patient in the lithotomy position. Some styles have large leggings incorporated into it to cover the legs in stirrups. It may have one or two openings to accommodate the perineum or rectum.

Laparoscopy Sheet. A laparoscopy sheet is a combination laparotomy and perineal sheet. It is used for gynecologic laparoscopy in lithotomy or combined abdominoperineal resection with the patient in the lithotomy position. The fenestrations are located on the abdominal portion and the perineum.

Split Sheet. The split sheet is the same size as a laparotomy sheet. Rather than the sheet being fenestrated, one end is cut longitudinally up the middle at least one third the length of the sheet to form two free ends (tails). The upper end of this split may be in the shape of a U.

Minor Sheet. The minor sheet is 36 × 45 inches (91 × 114 cm). It has many uses. Wrapped around an extremity, it permits the extremity to remain on the sterile field for manipulation during the surgical procedure. It is used under an arm to cover an arm board for shoulder, axillary, arm, and hand procedures.

Medium Sheet. The medium sheet is about 36 × 72 inches (91 × 183 cm). It is used to drape under legs, as an added protection above or below the surgical area, and for draping areas in which a fenestrated sheet cannot be used.

Single Sheet. The single sheet is 108 × 72 inches (274 × 183 cm). Folded lengthwise, it is placed above the sterile field to shield off the anesthesia provider and anesthesia machine or other equipment near the patient's head or sterile field. A single sheet also is used to cover the patient and OR bed below the surgical area around the face.

Leggings. Leg drapes are supplied in pairs to cover the legs of a patient in the lithotomy position. A rectangle, approximately 36 × 72 inches (91 × 183 cm), is closed on two sides to form a tent like pocket.

Basic principles regarding draping are as follows:

- Place drapes on a dry area. The area around or under the patient may become damp from solutions used for skin preparation. The circulating nurse removes damp items or covers the area to provide a dry field on which to lay sterile drapes.
- Allow sufficient time to permit careful application.
- Allow sufficient space to observe sterile technique. Do not reach across a nonsterile surface.
- Handle drapes as little as possible.
- Never reach across the OR bed to drape the opposite side; go around it.
- Take towels and towel clips, if used, to the side of the OR bed from which the surgeon is going to apply them before handing them up.
- Carry folded drapes to the OR bed. Watch the front of the sterile gown; it may bulge and touch the nonsterile OR bed or blanket on the patient. Stand well back from the nonsterile OR bed.

The draping procedure is as follows:

1. The surgeon places a drape under the head while the circulating nurse or assistant elevates the head. This drape consists of an open towel placed on a medium sheet. The center of the towel edge is 2 inches (5 cm) in from the center of the sheet edge. The towel is drawn up on each side of the face, over the forehead or at the hairline, and fastened with a small non-perforating towel clip. This leaves the desired amount of the face exposed.
2. Hand up three additional towels and four towel clips. These towels frame the surgical site.

3. Place a medium sheet just below the site. This sheet must overlap the one under the head.
4. A fenestrated drape may be placed to complete draping.
5. Cover the remainder of the foot of the OR bed, as necessary, with a single sheet. If the patient is receiving inhalation anesthesia, use a minor sheet instead of a towel on a medium sheet for the first drape under the head. A minor sheet is large enough to draw up on each side of the face and to enclose the endotracheal tube and oropharyngeal monitoring probes from the anesthesia machine for a considerable distance, thus keeping them from contaminating the sterile field. If the surgical procedure on the face is unilateral, the anesthesia provider may sit along the unaffected side, near the patient's head, with the anesthesia screen placed on the same side of the OR bed. Skin staples or sutures may be used to affix towels around the contours of the face and neck of the patient under general anesthesia. Each staple or stitch overlaps the skin and edge of the drape. Eye. After skin preparation, the unaffected eye is protected by covering it with a sterile eye pad before draping the patient. The draping procedure is as follows:

1. The surgeon places two towels and a medium sheet under the head while the circulating nurse holds the head up, as described for a face drape. One towel is drawn up around the head, exposing only the eyebrow and affected eye, and fastened with a clip without pressure on the eyes.
2. Hand up four towels and towel clips to isolate the affected eye. Some surgeons prefer a self-adhering aperture drape.
3. Cover the patient and remainder of the OR bed below the surgical area with a single sheet. If local anesthetic will be administered, the drapes are raised off of the patient's nose and mouth to permit free breathing. A Mayo stand or anesthesia screen positioned over the lower face before the draping is one method used to elevate the drapes. Oxygen, 6 to 8 L/min, can be supplied under the drapes via tube or nasal cannula. Take extreme caution that oxygen does not build up under the drapes. An ignition source such as a cautery or laser could spark a fire beneath the drapes.

For a microsurgical procedure, a sterile, padded, U-shaped steel wrist rest for the surgeon and assistant is fastened to the head of the OR bed. Towels are put around the patient's head before the rest of the facial and body draping is completed. If irrigation will be used, a plastic fenestrated drape is placed over the four towels to keep them dry if an aperture drape is not preferred. Ear. The basic draping procedure is the same as for draping a face or eye, except that only the affected ear is exposed. The head is turned toward the unaffected side. Oxygen can be supplied under the drapes, as previously described. The anesthesia provider is usually positioned at the side of the OR bed near the patient's face. Chest and Breast. While the arm is still being held up by the assistant after skin preparation:

1. Place a minor sheet on an arm board, under the patient's arm, extending the sheet under the side of the chest and shoulder. The prepped arm is lowered to the sterile draped

arm board. The distal portion of the arm may be encased in sterile stockinette so that the arm can be manipulated during the surgical procedure.

2. The patient's body is draped with a sterile medium sheet.
3. Hand up towels and towel clips; five or six are necessary.
4. Apply a breast sheet so that the axilla is exposed for anticipated axillary dissection.

Many breast procedures require both breasts to be prepped and exposed for comparison during the procedure. Shoulder. While the arm is still being held up by the assistant after skin preparation:

1. Place medium sheets over the chest and under the arm.
2. Place a minor sheet under the shoulder and side of the chest.
3. The surgeon outlines the surgical site with towels and secures them with clips.
4. Place a minor sheet over the patient's chest, covering the neck. Keep this sheet even with the edge of the towel that borders the surgical site laterally.
5. Wrap the arm in a minor sheet or encase it in a sterile stockinette, and secure it with a sterile gauze or elastic bandage. At this point, a sterile team member relieves the unsterile person who has been holding the arm.
6. Place a medium sheet above the area, and secure these sheets together with towel clips.
7. A laparotomy or breast sheet may be used. Pull the arm through the opening. Or a single sheet may be placed above the area, and the foot of the OR bed is covered with a medium sheet. Elbow. While the arm is still being held up by the unsterile assistant after skin preparation:

1. Place a sterile medium sheet across the chest and under the arm, up to the axilla.
2. The surgeon defines the surgical area on the upper (proximal) arm by placing a towel around the upper arm and securing it with a towel clip.
3. Wrap the hand in a sterile towel. At this point, a sterile team member relieves the unsterile person who has been holding the arm by grasping the wrapped hand, maintaining the arm in an elevated position.
4. The hand is grasped with a sterile stockinette, which is pulled down over the entire arm toward the axilla, over the surgical site. An elastic bandage is wrapped around the arm starting at the distal end (hand) to the proximal area (axilla).
5. Place a medium sheet across the chest, on top of the arm, even with the towel on the upper arm and covering it. Secure this sheet around the arm with a towel clip.
6. An extremity sheet is drawn over the hand and the arm. The extremity sheet is opened in its entirety across the patient's body. Hand. While the arm and hand are still being held up after skin preparation:

1. Place an impervious minor sheet, folded in half, on the extremity table.
2. The surgeon places a towel around the lower arm, limiting the exposed area to the affected hand, and secures it with a towel clip.

3. Pull a stockinette over the hand and up over the length of the arm. At this stage in draping, the unsterile person is relieved of holding the arm. The draped arm is laid on the draped extremity table.
4. Place a minor sheet across the extremity table just above the surgical site.
5. Place an extremity sheet over the hand. Do not drop folds below the level of the arm board. Open the sheet across the patient's body toward the feet first. Attach the top end to the IV poles at the head of the bed. Perineum. With the patient in the lithotomy position for a genital, vaginal, or rectal procedure:

1. Place a medium sheet under the buttocks. The circulating nurse can grasp the underside and assist in placement. With the patient's legs elevated in stirrups, this drape hangs below the level of the OR bed and covers the lowered section of the OR bed.
2. Slide leggings over each leg, protecting the gloved hands in the folded cuffs.
3. The anus is covered if it is not part of the surgical site. An adhesive towel drape may be used for this purpose.
4. Place a medium sheet across the abdomen, from the level of the pubis, extending over the anesthesia screen or attached to IV poles at the head of the bed.
5. A fenestrated perineal sheet may be used rather than a medium sheet over the abdomen. To use a perineal sheet with built-in leggings, hand one end of the sheet to the assistant, opening out folds, and draw the leggings over the feet and legs simultaneously. The hands are kept on the outside of the sheet to avoid contaminating the gloves and gown. Hip. The patient is in a lateral position. If the leg will be manipulated during the surgical procedure, while the leg is still being held up after skin preparation:

1. Place a medium sheet on the OR bed under the leg, up to the buttock.
2. Place another medium sheet on the OR bed, overlapping the first one, to cover the unaffected leg. Some surgeons prefer to use an incise sheet or a lower extremity stockinette.
3. The surgeon wraps the foot and leg with an elastic bandage, covering the stockinette. The leg, held up to this point, is laid on the OR bed.
4. Place a minor sheet lengthwise of the OR bed on each side of the exposed area, even with the skin. The sheets under the leg and above the site do not overlap. Some surgeons prefer to draw the leg through the opening of a hip sheet or place a split sheet under the leg with the tails crossed over it toward the patient's head.
5. Place a medium sheet above the exposed area. Secure these last three sheets with towel clips.
6. Place a single sheet above the surgical area and over the anesthesia screen. If manipulation of the leg is not necessary during the surgical procedure, drape the same as for a laparotomy, using a hip sheet instead of a laparotomy sheet.

Before the First Surgical Procedure of the Day

The following housekeeping duties should be done before bringing sterile surgical supplies into the room for the first case of the day:

1. Remove unnecessary tables and equipment from the room. Arrange the appropriate furniture in an organized manner away from the traffic pattern. Some head and neck procedures require the OR bed to be oriented in a sideways direction to provide working space for the team. The anesthesia provider might be positioned at the patient's side.
2. Damp-dust from the top down (with a facility-approved sporicidal disinfectant solution and lint-free cloth) the overhead operating light, articulated arms, furniture, flat surfaces, and all portable or mounted equipment. Avoid dry-dusting because this sets dust aloft. Start at higher surfaces, and work down to lower levels because dust may fall from higher areas. "High touch" areas such as knobs, handles, and keyboards must be cleaned after every patient.
3. Damp-dust the tops and rims of the sterilizer and/or washer sterilizer and the countertops in the sub sterile room adjacent to the OR.
4. Visually inspect the room for dirt and debris. The floor may need to be damp-mopped.

Getting the Room Ready for the Next Patient

The cleaning procedures described provide adequate decontamination and terminal sterilization after any surgical procedure. With a well-coordinated team, minimal turnover time between surgical procedures can be accomplished. In an average time of 10 to 15 minutes, the room will be ready for the next patient. The turnover time includes cleaning up after one procedure and setting up for the next procedure. Additional equipment brought into the room for the next patient should be damp-dusted with sporicidal disinfectant before entering the OR where sterile supplies are opened. Individual Patient Setups Each patient has a right to individual supplies prepared just for him or her. Sterile supplies should not be opened until they are ready to be used. Case cart systems and the use of custom packs eliminate the need for preparing the sterile field several hours ahead of the patient's arrival. The scrub person, working with an efficient circulating nurse, should have time to set up the instrument table immediately before each surgical procedure. There is no arbitrary life span of a setup table once it is open and prepared as a sterile field for a patient. Sterility is event related, not time related. The sterile table must be under surveillance at all times. Unless it is under constant surveillance, sterility of any setup cannot be guaranteed. If a scheduled surgical procedure is delayed and a sterile setup has not been contaminated by the patient's presence in the room, the setup may remain open, under surveillance by someone in the room, with the doors closed. Taping the door shut has no assurance of sterility of the setup. The setup should be used as close to the time of preparation as possible. If a patient is taken into the OR and for some reason the surgical procedure is canceled before the procedure has begun, the tables should be torn down and the room cleaned as if the surgical procedure had taken place. The setup is considered potentially contaminated and may not be saved for another patient. Disposable items may be useful for the clinical educator in the department during orientation and education sessions.

Aseptic Technique

It is impossible to exclude all microorganisms from the environment, but for the safety of both patients and personnel, every effort is made to minimize and control these microorganisms.

Microorganisms are present in the air and on animate and inanimate objects at all times. To effectively apply the principles of asepsis, environmental control, and sterile techniques discussed in this chapter, the meaning of terms related to aseptic technique must first be understood. Therefore the basis of prevention is the knowledge of causative agents and their control, as well as the principles of aseptic and sterile techniques. The methods by which microbial contamination is contained in the environment are referred to as aseptic technique. The OR in the restricted area is aseptic at best because the room and air cannot ever be 100% free of microbial content. Some key elements of asepsis include the following facts:

- Aseptic technique is sometimes referred to as clean technique.
- Items have been cleaned and decontaminated so they are safe to handle with clean, bare hands.
- Items in use in patient care are handled with examination gloves for the protection of both the caregiver and the patient.
- Items have been cleaned, decontaminated, disinfected, or terminally sterilized without a wrapper, and stored in a clean, dry place.
- Items may start out sterile but are not maintained or used under sterile conditions. Skin preps may be packaged sterile, but skin cannot be sterilized. The process is aseptic.
- Contamination is contained. Extraneous contamination is avoided.
- Items are set up on clean towels or drapes and used with examination gloves.
- Items are not sterile or maintained sterile during use. Extraneous contamination is avoided.
- Disposable items are not cleaned and reused for another patient.
- Items are classified as semi critical or noncritical by Spaulding's classification of the importance of patient care items.
- Items can be used outside the confines of the restricted area.
- Items are used on intact skin or mucous membranes.
- Items are not used when the patient's vascular system will be entered.

Sterile Technique

Sterile technique incorporates many processes associated with asepsis but to a higher, more controlled degree. In sterile technique all microorganisms must be maintained at an irreducible minimum meaning as low as absolutely possible. Keep in mind that a sealed sterile package must be opened at some point during patient care to use the contents. That means opening the package to the room air. Even in the restricted area the room air has a microbial count that we cannot eliminate. We try as hard as possible to minimize the room traffic and maintain environmental controls to maintain the sterile field. Essential elements of sterile technique include the following factors:

- Items are used only in a sterile field in the restricted area.
- Items are used by sterile team members wearing appropriate sterile attire.
- Items are used in areas of the patient's body where the site has been prepped.
- Items may be used in invasive surgery.

- Items may be used in body areas with non-intact skin and membranes and may enter the patient's vascular system.
- Items are classified as critical according to Spaulding's level of importance of patient care items.
- Items have been cleaned, decontaminated, and packaged before sterilization.
- Items processed by sterilization are stored wrapped and remain so until use by a sterile team member.
- Items are for individual patient use only. Reusable items can be reprocessed and resterilized for use with another patient. Disposable items are discarded after use. If opened and unused, disposable items are discarded.
- Items that become contaminated are discarded and replaced immediately.

Principles of Sterile Technique Sterile technique is the foundation of modern surgery

The patient is the center of the sterile field, which includes the personnel wearing sterile attire and the areas of the patient, operating bed, and furniture that are covered with sterile drapes. The level of the surgical site is the central focus of the level of the sterile field. Strict adherence to the recommended practices of sterile technique reflects the surgical conscience of the perioperative team and is mandatory for the safety of the patient and personnel in the environment. The principles of sterile technique are applied under the following conditions:

- In preparation for an invasive procedure by sterilization of necessary materials and supplies
- In preparation of the sterile team to handle sterile supplies and intimately contact the surgical site by scrubbing, gowning, and gloving
- In the creation and maintenance of the sterile field, including skin preparation and draping of the patient
- In the maintenance of sterility throughout the entire surgical procedure; breaches in sterility are remedied immediately
- In terminal decontamination, disinfection, and sterilization at the conclusion of the surgical procedure only sterile items are used within the sterile field Items such as sterile instrument sets, drapes, sponges, and basins are obtained from the sterile core. Only if absolutely necessary, items such as unwrapped instruments may be steam sterilized immediately before the surgical procedure and taken directly from the sterilizer to the sterile field. This rapid steam sterilization method is referred to as flashing and should be used only if no other alternative is available. Most facilities keep a record of each time flashing is used. Every person who dispenses a sterile item to the field must be sure of its initial and continued sterility until used. Proper packaging, sterilizing, delivery to the field, and handling should provide such assurance. If there is any doubt about the sterility of any item, it should be considered unsterile and therefore should not be used. The phrase "when in doubt, throw it out" applies to this situation. Examples of questionably sterile items include, but are not limited to, the following situations:

- If a sterilized package is found in a contaminated area (e.g., the general unsterile workroom, the locker room)
- If uncertain about the actual timing or operation of the sterilizer. Items processed in a suspect load are considered unsterile. Other items processed in that load may need to be recalled to the processing department for reprocessing (e.g., the external process monitor has irregular color changes).
- If an unsterile person or object comes into close contact with a sterile table and vice versa
- If a sterile table or unwrapped sterile items are not under constant observation (e.g., a sterile field under a table cover is not directly visible)
- If the integrity of the packaging material is not intact
- If a sterile package wrapped in a material other than plastic or another moisture-resistant barrier becomes damp or wet. Humidity in the storage area or moisture on hands may seep into the package.
- If a sterile package wrapped in a previous woven material drops to the floor or other area of questionable cleanliness. These materials allow the implosion of air into the package. A dropped package is considered contaminated on the outside. If the wrapper is impervious and the area of contact is dry and intact, the outer wrapper is removed and item may be transferred to the sterile field. Packages that have been dropped on the floor should not be put back into sterile storage.

Sterile Personnel Are Gowned and Gloved

Gowns are considered sterile only from the chest to the level of the sterile field in the front, and from 2 inches above the elbows to the cuffs on the sleeves. When wearing a gown, only the area that can be seen in front down to the level of the sterile field should be considered sterile. Keep in mind that the level of the patient's surgical site actually establishes the level of the sterile field Usually this does not extend below waist level. The following practices are observed:

Self-gowning and gloving should be done from a separate sterile surface to avoid dripping water onto sterile supplies or a sterile table. Closed gloving technique is preferred for the person who is establishing the sterile field and gowning and gloving others.

The stockinette cuffs of the gown are enclosed beneath sterile gloves. The stockinette is absorbent and retains moisture; therefore this part of the gown does not provide a microbial barrier and permits the transfer of microorganisms. Once the cuff has been contained within the closed glove, it should not be pulled back over the hand for any regloving procedure because it is contaminated. Regloving should be performed using the open assisted gloving technique. Another sterile team member can assist with this process as necessary.

- Sterile people must keep their hands in sight at all times and at or above waist level or the level of the sterile field.
- Hands are kept away from the face and the elbows are kept close to the sides. The hands are never folded under the arms because of perspiration in the axillary region. The neckline, shoulders, and back also may become contaminated with perspiration. The back of the gown is not under constant observation and therefore is considered contaminated.
- Sterile people are aware of the height of team members in relation to each other and the sterile field. Changing levels at the sterile field is avoided. All team members must be

aware of spatial relationships at all times. The gown is considered sterile only down to the highest level of the sterile working field. If a sterile person must stand on a platform to reach the surgical site, the standing platform should be positioned before this person steps up to the draped area. Sterile personnel should sit only when the entire

- Tables Are Sterile Only at Table Level Because tables are sterile only at table level, OR personnel must adhere to the following procedures:
- Only the top of a sterile, draped table is considered sterile. The edges and sides of the drape extending below table level are considered contaminated. The Mayo stand, when covered by a sterile drape, may be placed over the sterile field. Minimal contact is had with the underside of the Mayo drape.
- Anything falling or extending over the table or OR bed edge, such as a piece of suture or suction tip, is contaminated. The scrub person does not touch the part hanging below the level of the established sterile field.
- When unfolding or applying a sterile drape, it is unfolded away from the sterile person and the part that drops below the table surface is not brought back up to table level. Once placed, the drape is not moved or shifted.
- Cords, tubing, and other materials are secured on the sterile field with a non-perforating clip to prevent them from sliding over the edge of the operating bed.
- Sterile Personnel Touch Only Sterile Items or Areas; Unsterile Personnel Touch Only Unsterile Items or Areas
- Sterile team members maintain contact with the sterile field by means of sterile gowns and gloves.
- The unsterile circulating nurse does not directly contact the sterile field.
- Supplies are brought to sterile team members and opened by the circulating nurse using aseptic technique. The circulating nurse ensures a sterile transfer to the sterile field. Only sterile items touch sterile surfaces.
- Unsterile Personnel Avoid Reaching Over the Sterile Field; Sterile Personnel Avoid Leaning Over an Unsterile Area
- The unsterile circulating nurse never reaches over a sterile field to transfer sterile items.
- The circulating nurse holds only the lip of the bottle over the basin when pouring solution into a sterile basin to avoid reaching over a sterile area (Fig. 15-5). He or she should avoid making contact with the bottle and the basin and avoid splashing solutions. The entire contents of the bottle should be dispensed in one pouring motion. If some of the solution is not poured, it is considered contaminated and may not be recapped for later use or dispensed to another sterile basin. Once the cap is off the solution is considered contaminated.
- The unsterile solution may be saved for cleaning the patient's skin after the incision is dressed and the surgical site is undraped.
- The scrub person sets basins or medicine cups to be filled at the edge of the sterile table; the circulating nurse stands a safe distance away from the edge of the table to fill them. Medications are drawn up in a syringe, the needle is removed, and the drug is dispensed to the cup without aerosolization. It is unsafe to draw up a drug if another person is

holding the bottle. This is no different from recapping a needle by hand. It is not advised to dispense the drug via syringe and needle because the needle could cause an aerosolization of a drug to which a team member might be sensitive.

- The team uses sterile light handles for manipulation and adjustment of the surgical lights. Light handles should not be placed until after the patient is fully draped.
- The surgeon or team member steps away from the sterile field to have perspiration removed from the brow.
- The scrub person drapes an unsterile table by: Unfolded drape. Placing the drape over the unsterile surface nearest himself or herself first and carefully completing the coverage of the far side. This protects the front of the gown from coming into contact with the unsterile surface being covered. Gloved hands are protected by cuffing a drape over them.
- Folded drape. Unfolding toward self-first to protect the gown when working with a folded drape through). When completing the coverage, the rest of the drape is unfolded away from self.
- The scrub person stands back from the unsterile table when draping it to avoid leaning over an unsterile area. The Edges of Anything That Encloses Sterile Contents Are Considered Unsterile The boundaries between sterile and unsterile areas are not always rigidly defined (e.g., the edges of wrappers on sterile packages and the caps on solution bottles). The following precautions should be taken:
- Sterile Areas Are Continuously Kept in View Inadvertent contamination of sterile areas must be readily visible. To ensure this principle, the following steps must be taken:
- Sterile personnel face sterile areas.
- Someone must remain in the room to maintain vigilance when sterile packs are opened in a room or a sterile field is set up. Sterility cannot be ensured without direct observation. An unguarded sterile field should be considered contaminated.
- Sterile Personnel Keep Well Within the Sterile Area Sterile persons allow a wide margin of safety when passing unsterile areas and observe the following rules:
- Sterile personnel stand back at a safe distance from the OR bed when draping the patient.
- Sterile personnel pass each other back to back at a 360-degree turn (Fig. 15-10).
- Sterile personnel turn their backs to an unsterile person or area when passing.
- Sterile personnel face a sterile area to pass it.
- Sterile personnel ask an unsterile individual to step aside rather than risk contamination.
- Sterile personnel stay within the sterile field. They do not walk around or go outside the room.
- Movement within and around a sterile area is kept to a minimum to avoid contamination of sterile items or personnel. Sterile Personnel Keep Contact with Sterile Areas to a Minimum To keep contact with the sterile areas to a minimum, sterile personnel observe the following rules:
- Sterile personnel do not lean on sterile tables or the draped patient. Leaning on the patient can cause injury to tissues and structures.

- Sitting or leaning against an unsterile surface is a break in technique. If the sterile team sits to operate, they do so without proximity to unsterile areas.
- Unsterile Personnel Avoid Sterile Areas Unsterile personnel maintain an awareness of sterile, unsterile, clean, and contaminated areas and their proximity to each. They must be aware of their closeness to the sterile field. A wide margin of safety must be maintained when passing sterile areas by observing the following rules:
    - Unsterile personnel maintain a distance of at least 1 foot (30 cm) from any area of the sterile field.
    - Unsterile personnel face and observe a sterile area when passing it to be sure they do not touch it.
    - Unsterile personnel never walk between two sterile areas (e.g., between sterile instrument tables).

The integrity of a sterile package and the appearance of the process monitor must be checked for sterility just before opening. To ensure sterility, the following precautions should be taken:

- Sterile packages are laid only on dry surfaces.
- If a sterile package wrapped in absorbent material becomes damp or wet, it is discarded or wrapped in fresh wrappers

Learning outcome 3: Wind-Up Perioperation Theatre Clinical Services

Introduction to the learning outcome

This learning outcome involves re-arranging the Operation theatre, documentation of patients' condition, specimen handling, decontamination, types of wastes in Operation Theatre and respective disposal methods and last offices services in Operation Theatre

1.2.3 Performance Standard

1. Operation room is reorganized for the next procedure in accordance to Standard Operation Procedures
2. Documentation is carried out according to work place policy and WHO
3. Waste is disposed with due regard to environment protection regulations

1.2.4 Information Sheet

Definition of key terms

PACU - Post Anesthesia Care Unit

Content/methods/illustrations

Re-arranging the Operation Theatre

After the Surgical Procedure Is Complete the scrub person will apply the dressing after the incision is cleaned and dried. The drapes will be rolled off the patient as a sterile team member holds the dressing in place. Clean the surrounding area of the patient's body before securing the dressing(s) over the surgical wound and the surgical drainage systems. Put a clean, warm gown and blanket on the patient. Place the safety belt over top of the blanket. It needs to be seen at all times. Open the neck and back closures of the surgeon's and assistants' gowns so they can remove them without contaminating themselves. Have a patient care assistant bring in the clean transport cart or bed. Check the patient's name on the stretcher or bed if it is the procedure to return the patient to the same one used for transport to the OR suite. Align the cart with the OR bed, and lock the wheels. Remove the leg safety belt. A lifting frame or patient roller is a great help in moving unconscious and obese patients. The Davis patient roller consists of a series of rollers that are mounted in a frame long enough to accommodate the trunk of an adult patient. The edge of the roller is placed under the lift sheet and the patient's side. With the patient's head and feet supported by separate team members, the lift sheet is pulled and the patient is rolled onto the stretcher or bed. A minimum of four people is required to safely move the patient with this device: one to lift the head and manage the airway (usually the anesthesia provider), one to lift the feet and control the Foley catheter, one beside the stretcher or bed to pull, and one beside the patient to lift him or her from the OR bed. The action of all four people should be synchronized, and the count of three is called by the person controlling the head, usually the anesthesia provider. The following precautions should be taken in lifting or rolling an unconscious patient:

1. Protect the IV, drains, and urinary drainage bag. Secure IV solution bags on an IV pole. It is preferable to attach the IV pole near the foot end of the stretcher or bed, where there is less danger of injury to the patient if the bag or IV pole should fall or break.
2. Use the lift sheet to support the arms at the sides so the arms do not dangle.
3. The anesthesia provider guards the head and neck from injury and calls the count for the move.
4. Lift or roll the patient gently and slowly to avoid circulatory depression. After the patient is safely on the transport cart, remove the lift sheet by logrolling the patient gently from side to side. Brush burns result if the fabric is pulled from under the patient. Place the patient in a comfortable position that is most conducive to the maintenance of respiration and circulation. The appropriate position may vary with the type of surgical procedure; usually the patient should:

- Be supine after a laparotomy, with the head of the stretcher elevated 15 degrees, especially if he or she is still intubated
- Be semi prone after a tonsillectomy, for drainage, if extubated
- Be lateral, on the affected side, after transthoracic surgical procedures, thus splinting the side, usually extubated
- Have the affected extremity supported on a pillow Raise the side rails before the patient is transported out of the OR. Be sure the completed chart and proper records accompany the patient, and send other supplies as indicated (tracheostomy obturator). Help transport the patient to the PACU or patient care unit. Patient Transfer from the Operating Room During transport, the patient should be constantly observed by someone familiar with his or her condition. If the patient has had local anesthesia, the circulating nurse and transporter may accompany the patient during the return to an ambulatory unit. Hand-Off Report to PACU Nurse If the patient has had general anesthesia, the anesthesia provider and circulating nurse should accompany him or her during transport to the PACU, where they give a verbal hand-off report to the PACU nurse. This postoperative report is important for the continuity of care. A concise verbal report from the anesthesia provider and/or circulating nurse includes the following:

• Name and age of the patient
• Type of surgical procedure and name of the surgeon
• Type of anesthesia and name of the anesthesia provider
• Vital signs (baseline, preoperative, and intraoperative), including current body temperature. Body weight should be available in kilograms.

Types and locations of drains, packing, and dressings. Closure medium should be included.

• Preoperative level of consciousness and current status
• Medications given preoperatively and intraoperatively, as well as those regularly taken by prescription or self-medication
• Medical and surgical history
• Allergies and responses to allergens, substance sensitivity
• Positioning on the OR bed and devices attached to the skin
• Complications during the surgical procedure
• Intake and output, including IV fluids, blood, and urinary output
• Location of the waiting family or significant others • Special considerations: • Sensory and/or physical impairments; eyewear, hearing aids, dentures, or other personal property brought to the OR is returned to the patient when the level of consciousness is appropriate • Language barrier and level of understanding • Use of tobacco, alcohol, and/or addictive drugs • Orders such as “no code,” “do not resuscitate (DNR),” or “do not attempt resuscitation”

Sponge, Sharps, and Instrument Counts The counting of sponges, needles and other sharps, and instruments has been mentioned throughout this chapter. These supplies are crucial to the surgical procedure and must be accounted for throughout every procedure, regardless of size or function. Items are counted before and after use. The types and numbers of sponges, needles and other sharps, and instruments vary for each surgical procedure. The process of counting gives the circulating nurse and the scrub person the opportunity to look at all the supplies and the entire instrument set before the procedure begins. Missing or defective items can be obtained or replaced without interrupting the continuity of the surgery. Accountability for counts is a professional responsibility that rests primarily on the scrub person and the circulating nurse. The surgeon and patient rely on the accuracy of this accountability by the team. There are several reasons why it is important for the scrub person and circulating nurse to count and be accountable for all items used during the procedure. Counts are performed for patient and personnel safety, infection control, and inventory purposes. A retained surgical item in the surgical site after closure is a possible cause for a lawsuit after a surgical procedure. A retained foreign object made of woven textile is referred to as a gossypiboma or a textiloma. A foreign body unintentionally left in a patient can be the source of wound infection or disruption.

The longer the object remains in the body, the more it incorporates ingrowth of tissue. An abscess can form, and fistulas may develop between organs. The foreign body reaction may be immediate or may be delayed for years. Diagnosis is sometimes difficult and costly, and removal of the object usually requires major surgery. The literature reports the removal of some retained sponges through laparoscopic surgery if they are discovered before adhesions develop.

A contaminated sponge or needle that is unaccounted for at the close of the procedure could also inadvertently come into contact with the personnel who clean the room, process instruments, launder the linens, or transport the trash. Blood or body fluids are sources of pathogens such as human immunodeficiency virus (HIV) or hepatitis B virus (HBV). A surgical pattie (cottonoid) that has become saturated with clear cerebrospinal fluid during a neurologic case may also be contaminated with Creutzfeldt-Jakob disease (CJD). Inventory control is monitored by accounting for the instrument set in its entirety. Counting ensures that expensive instruments such as towel clips and scissors are not accidentally thrown away or discarded with the drapes. Injury to laundry and housekeeping/environmental services personnel by the contaminated sharp edges of surgical instruments, blades, and needles is a risk. Surgical instruments also can cause major damage to equipment in the laundry services. Unfortunately, some facilities have felt a need to install metal detectors in the trash and soiled linen areas to monitor for missing instruments. Counting Procedures a counting procedure is a method of accounting for items put on the sterile table for use during the surgical procedure. Sponges, sharps, and instruments should be counted and/or accounted for on all surgical procedures. This includes any materials introduced into the patient during the procedure, such as rectal or vaginal packs or sterile towels used to pack off or retain viscera. Inadvertent items such as injection needle caps, cautery tip cleaners, clip cartridges, or suture reels could become retained objects and should be accounted for in their entirety. Items used during organ procurement procedures should be accounted for in their entirety in the same manner as for any surgical procedure. Respect for the donor should be as important to the surgical team as respect for any patient. Initial Count When the Instrument Tray Is Assembled. The person who assembles and wraps items for sterilization will count them in standardized multiple units and initial the baseline set contents. Most facilities enclose a copy of this tray inventory count sheet in the instrument set or attach a copy to the outer wrapper. The effects of enclosing a tray inventory sheet inside the set raises questions of potential exposure of toner and foreign body reaction in the patient. Methods for attaching the tray sheet to the outside of the set should be explored if this is a concern. In commercially prepackaged sterile items (e.g., sponges, disposable towels), this count is performed by the manufacturer and affixed to the outside of the package. Baseline count during Setup for the Surgical Procedure. The scrub person and the circulating nurse together count all items before the surgical procedure begins and during the surgical procedure as each additional package is opened and added to the sterile field. These initial counts provide the baseline for subsequent counts. Any item intentionally placed in the wound, such as towels, is recorded. A useful method for counting is as follows:

1. As the scrub person touches each item, he or she and the circulating nurse number each item aloud until all items are counted.

Reversal agents and their roles

- Flumazenil, reverses the effects of benzodiazepines
- Naloxone, reverses the effects of opioids
- Neostigmine, helps reverse the effects of non-depolarizing muscle relaxants
- Sugammadex, new agent that is designed to bind Rocuronium therefore terminating its action

The following are activities/responsibilities of the scrub person at the end of the case:

1. Push the Mayo stand and instrument table away from the OR bed as soon as the dressing is applied and the drapes are removed. Roll drapes off the patient from head to foot to prevent airborne contamination; do not pull them off.
2. Check drapes for towel clips, instruments, and other items. Be sure that no equipment is discarded with disposable drapes or sent to the laundry. Disposable drapes are placed in a red biohazard container for disposal. Soiled drapes, whether disposable or reusable, should be handled as little as possible and with minimum agitation to prevent gross microbial contamination of air by dispersal of lint and debris.
3. Discard soiled sponges, other biologically contaminated waste, and disposable items in red biohazard containers. Discard unused sponges, nonwoven drapes, and other disposable waste into the main trash.
4. Dispose of sharp items safely. Special care should be taken in handling all knife blades, trocars, burrs and bits, surgical needles, and needles used for injection or aspiration. Remove the tip from the ESU handle (pencil). A self-closing adhesive pad or box designed for this purpose is the safest device to use. A safe disposal procedure should be implemented and sustained. Place these items in an appropriate rigid, puncture-resistant container for safe disposal to prevent injury and potential risk of contamination. The primary cause of accidental cuts and punctures to personnel, both inside and outside the OR, is disposal of surgical sharps at the end of the surgical procedure. Adherence to standardized systems designed specifically for safe handling and disposal of sharps prevents virtually all accidental cuts, punctures, and lacerations. Unused suture packets are discarded.
5. Basins and trays too large for the case cart are put into plastic bags for transport to the decontamination area. The Mayo tray may be included at some facilities. Place these on the lower shelf of the case cart.
6. The instruments are opened completely and placed into the wire mesh basket with all box locks spread apart. Blood, tissue, bone, and any other gross debris are removed from instruments during the case as much as possible. All instruments, used and unused, must be decontaminated, terminally sterilized, or undergo high-level disinfection (as appropriate) before they are processed for reuse. Instruments should be presoaked and/or pre-rinsed before processing in a washer-sterilizer or decontaminator. Some facilities have the scrub person spray enzymatic foam over the instruments to start the cleaning process. Any biologic material remaining on instruments is more difficult to remove after the instruments have been heat-sterilized because the material becomes baked on them. The biologic debris inhibits sterilization and disinfection processes. a. Remove knife blades from handles using a heavy hemostat; never use fingers. Using a needle holder can cause the jaws of the instrument to become misaligned. Point the blade toward the table and away from the field and other people in the area so that if it breaks or slips it will not fly across the room. b. Unloaded scalpel handles and other instruments with sharp tips or edges, such as scissors, should be placed in a container separate from the other instruments so they can be easily identified by the processing personnel. Do not put knife handles in an instrument tray with blades left on them. Other instruments designed for

replaceable cutting blades, such as dermatomes, should have blades removed; thereafter they may be handled with other instruments. Place reusable surgical needles, either on a needle rack or loose, into a perforated stainless steel box to be decontaminated and sterilized with the instruments.

7. Dispose of solutions and suction bottle contents in a flushing hopper connected to a sanitary sewer. Wear PPE to protect from splashes. Some facilities have closed system liquid waste disposal units that empty suction canisters and decontaminate the contents. Disposable suction units simplify disposal. Commercial substances can be added to liquid in the disposable canister to solidify or gel contents for solid waste disposal.

8. The scrub person removes gown and gloves before taking the case cart to the processing area. The gown is removed first before removing gloves. The circulating nurse unfastens neck and back closures. Protect arms and scrub clothes from the contaminated outside of the gown. The gown is turned inside out as it is removed to prevent contamination of the scrub suit. Discard the gown in a laundry hamper if it is reusable or in a trash receptacle if it is disposable. To remove gloves, use a glove-to-glove and then skin-to-skin technique to protect the hands from the contaminated outside of gloves. Turn gloves inside out as they are removed to contain the biologic contamination, and then discard them into a trash receptacle. Wash hands after removing gloves.

9. The case cart should be totally enclosed or covered in plastic before taking it to the decontamination and processing area. Fresh examination gloves are worn when transporting the case cart.

Room Turnover Activities by the Team After the patient leaves, the environmental service personnel should be available to perform room cleaning. Regardless of which member of the team performs them, specific functions should be performed to complete the between case room cleanup. The following personnel and areas are considered contaminated during and after the surgical procedure:

• Members of the sterile team, until they have discarded their gowns, gloves, caps, masks, and shoe covers. These items remain in the contaminated area; scrub clothes are changed if they are wet or contaminated.

All furniture, equipment, and the floor within and around the perimeter of the sterile field. If accidental spillage has occurred in other parts of the room, these areas are also considered contaminated.

All anesthesia equipment.

Stretchers used to transport patients and patient moving devices. These should be cleaned after each patient use. Clean examination gloves are worn to complete the room cleanup. Decontamination of the environment includes the following tasks starting at the highest point and working down:

Overhead operating light. Wipe overhead lights with a clean cloth that has been wetted with sporicidal disinfectant solution specifically intended to prevent clouding of the surface that can cause dullness and glare. Lights and overhead tracks become contaminated quickly and present a possible hazard from fallout of dust particles onto sterile surfaces or into wounds during surgical procedures.

Furniture. Wash horizontal surfaces of all tables and equipment, including the anesthesia machine, with a sporicidal disinfectant. Apply disinfectant from a squeeze-bottle dispenser, and wipe with a clean cloth or a disposable wipe that is changed frequently. Spray bottles can cause particles to become aerosolized and should be avoided.2 All surfaces of mattress, pads, and screw connections of the OR bed are included. Safety straps should be cleaned between patients. Velcro straps can be laundered according to the manufacturer's recommendations. Mobile furniture can be pushed through sporicidal disinfectant solution used for floor care to clean casters. Excess strands of loose suture should be removed from the wheels. Knobs, machine controls, keyboards, cabinet handles, and other frequently touched surfaces must be cleaned with sporicidal disinfectant between patients.

Anesthesia equipment. Most masks and anesthesia tubing are disposable. Any reusable anesthesia masks and tubing are cleaned and sterilized between patient uses. Some of this equipment can be steam-sterilized; if not, it may be sterilized by ethylene oxide gas and aerated before reuse. If this method is not available, items should be chemically sterilized according to the sterilant manufacturer's recommendations.

Laryngoscope blades and handles should be disassembled, thoroughly decontaminated, and disinfected. Any parts that can tolerate a sterilization process should be terminally sterilized.

Noncritical items, such as blood pressure cuffs, should be cleaned with an approved sporicidal disinfectant between patient uses.

Laundry. After all cleaning procedures have been completed, discard cleaning cloths or put into a laundry bag if they are not disposable. When all reusable woven fabric items, used and unused, have been placed inside the laundry bag, close it securely. To help protect laundry personnel, an alginate bag that dissolves in hot water may be used as the primary laundry bag or as a liner within a cloth bag. Transport reusable woven fabrics soiled with blood or body fluids in leak-proof bags.

Trash. Collect all trash in plastic or impervious bags, including disposable drapes and kick bucket and wastebasket liners. Bags should be sturdy to resist bursting or tearing during transport. Trash can be separated into biohazardous waste, noninfectious trash, and recyclable items. Separate receptacles for each type of trash should be available. Disposition of potentially infectious waste must comply with local, state, and/or federal leak-proof regulations for contamination control measures. Use appropriately labeled and color-coded bags for infectious waste, and use puncture-resistant containers for sharps.

Floors. Clean a perimeter of 3 to 4 feet in circumference of the surgical field between cases. This perimeter expands from the clean area to dirty area—in the direction of visible soilage. Hot water may hasten the biocidal/sporicidal action of the disinfectant agent but may also soften tile adhesive. Standing platforms (step stools) are considered part of the floor and should be cleaned between cases.

Mops. Fresh, clean mops are used with fresh Environmental Protection Agency (EPA)–registered sporicidal disinfectant solution. The floor can be flooded with fresh detergent-disinfectant solution. A fresh mop is used to apply solution and used to take up solution beginning in the cleanest area and working toward the dirtiest areas. After one-time use, remove mop heads and place in a laundry hamper with other contaminated reusable woven fabrics. Mop handles should be cleaned with disinfectant after use and stored in the housekeeping storage area until they are needed again. Use clean mops and sporicidal disinfectant solution for each cleanup procedure. Wet vacs can be used to clean the OR flooring. The wet vac unit should be cleaned and disinfected between uses.

Walls. If walls are splashed with blood or organic debris during the surgical procedure, wash those areas. Otherwise, walls are not considered contaminated and need not be washed between surgical procedures before processing to the required level of safety for patient use.

Decontamination combines mechanical or manual cleaning and a physical or chemical microbicidal process. Prerinsing, washing, rinsing, and disinfecting/sterilizing is done in the processing department to render the instrumentation safe for handling by CS personnel. This part of the process is referred to as thorough cleaning or return-to-use cleaning. After the formal decontamination process the instrumentation are assembled into sets by the CS technician and processed either by sterilization or disinfection (high level, intermediate level, or low level) for use by the OR staff in patient care. Methods and procedures for decontamination and disinfection vary according to the item to be processed and the recommended agent used in processing. Personal protective equipment should be worn while using chemical disinfecting agents.

Prerinsing/Presoaking

The purpose of prerinsing or presoaking is to prevent blood and debris from drying on instruments or to soften and remove dried blood and debris. The circulating nurse can prepare a basin or plastic bin (with a cover) of enzymatic solution for the scrub person. For a short immersion period, instruments may be presoaked in the following list of agents and solutions:

A triple proteolytic enzymatic detergent. Proteolytic enzymes dissolve blood and debris, and the detergent removes dissolved particulate from the surfaces of instruments, including otherwise inaccessible areas such as lumens.

An enzymatic agent diluted per manufacturer's instructions.

Sodium hypochlorite (chlorine bleach) is corrosive; however, it is used as a presoak in suspected or potential prion disease such as transmissible spongiform encephalopathy (TSE). Instruments should not be soaked in any chlorine compound for more than 1 hour and should not be autoclaved with chlorine solution because of the formation of toxic chlorine gas.

Soaking in a phenolic, guanidine thiocyanate, or sodium hydroxide is an alternative, but chlorine bleach is the most reliable in reducing the prion titer within 1 hour. These chemicals are extremely corrosive and are not appropriate for use on endoscopes. Prerinsing or presoaking instruments with enzymatic solution in the OR can make further processing more efficient. Instruments should not be cleaned in scrub sinks or utility sinks in the substerile room. Instruments in an enzymatic soak solution should be transported to the processing department in a covered container.

Manual Cleaning If a washer-sterilizer or washer-decontaminator is not available, instruments are washed by hand in the processing area. The purpose of manual cleaning is to remove residual blood and debris before terminal sterilization or high-level disinfection. Delicate and sharp instruments should be handled separately. Microsurgical and ophthalmic instruments should be cleaned and dried by hand; they are not put in a washer-sterilizer. Because moisture is conducive to corrosion, these instruments should not remain wet for long periods. Complex instruments require complete disassembly before cleaning. Other instruments require special care. For example, the outside surfaces of powered instruments must be cleaned, but these instruments cannot be immersed in liquid. Some powered equipment requires lubrication as part of the cleaning process. Personnel in the processing area wear PPE such as caps, gloves, waterproof aprons, and face shields to prevent accidental spray from contaminated body fluids and chemical cleaning solutions. The following steps should be observed when cleaning instruments manually:

Fill the washing sink with clean, warm water to which a noncorrosive, neutral pH, low-sudsing, free-rinsing detergent has been added.

Detergent should be compatible with the local water supply. The mineral content of water varies from one region to another; a water softener may be used in the system routinely to minimize mineral deposits. Regardless of the water content, the detergent should be anionic or nonionic and have a pH as close to neutral as possible. An alkaline detergent (pH more than 8.5) can stain instruments; an acidic detergent (pH less than 6) can corrode or pit them.

Proteolytic enzymatic detergents dissolve blood and protein and remove dissolved debris from crevices. These detergents are effective in a wide range of water qualities.

Liquid detergents are preferable because they disperse more completely than do solids. Dilute the concentration before contact with instruments to avoid corrosion and staining. Do not pour liquid or put solid detergents directly on instruments.

Wash instruments carefully to guard against splashing and creating aerosols.

a) Use a soft brush to clean serrations and box locks. A soft bristle toothbrush may be used to clean ophthalmic, microsurgical, and other delicate instruments. Keep instruments totally submerged while brushing to minimize aerosolizing microorganisms and chemicals.
b) Use a soft cloth to wipe surfaces. A non-fibrous cellulose sponge will prevent damage to delicate tips.
c) Remove bone, tissue, and other debris from cutting instruments.
d) Never scrub surfaces with abrasive agents such as steel wool, wire brushes, scouring pads, or powders. These agents will scratch and may remove the protective finish on metal, thus increasing the likelihood of corrosion. The finish on stainless steel instruments protects the base metal from oxidation.

3. Rinse instruments thoroughly in distilled or deionized water at the temperature recommended by the manufacturer. Some enzymes can be inactivated by extreme temperatures. The water should not exceed 140° F (60° C) to prevent burns of the skin. Inadequate rinsing can leave a surface residue that can stain instruments.

4. Load instruments into the appropriate trays for terminal sterilization or containers for high-level disinfection. a. Put instruments back into sterilization racks or replace the protective guards as appropriate. b. Arrange instruments that can be steam-sterilized in sterilizer trays for the washer-sterilizer or washer-decontaminator. The unwrapped tray is terminally steam-sterilized to make it safe for handling.

Perioperative Documentation

Specific care given in the perioperative environment should be documented on the patient's chart. Most facilities use a preprinted form with a standardized plan of care. Space is provided to add individualized patient needs and to document additional interventions. Data included in the record come from several patient care areas. A comprehensive checklist is included with the chart to assist the circulating nurse to determine whether all of the data

Incident Report When an accident or unusual incident occurs involving a patient, employee, or property in the facility, the factual details should be reported to the nurse manager and documented according to institutional policy.

Preliminary Preparations Preliminary preparations of the OR are completed by the circulating nurse and scrub person before each patient enters the OR. Assistance is provided by environmental service personnel. It is a cooperative effort. Clean, organized surroundings are part of total patient care. A visual inspection of the room and its contents should be performed by the team before bringing in supplies for a case. Keep in mind that the human eye cannot see microscopic contamination. The environmental sanitation activities described in this chapter are for the benefit of the patient's protection in the spirit of surgical conscience and should not be taken for granted as an unnecessary inconvenience for the staff. Preparation for the start of the day requires making sure the room is as clean as it can be.

Basic room contents should include the OR bed, anesthesia machine and supplies, electrosurgical unit (ESU), instrument table, preparation (prep) table, Mayo stand, suction apparatus, receptacles for biohazard and regular trash, and a linen hamper for reusable woven fabrics

Other tables and equipment are added as needed. Anything brought into the room for the procedure must be wiped down with an approved sporicidal disinfectant before put into use by the team.

Last offices services in Operation Theatre

The last offices, or laying out, is the procedures performed, usually by a nurse, to the body of a dead person shortly after death has been confirmed. They can vary between hospitals and between cultures.

•To prepare the deceased for the mortuary (a funeral home or morgue), respecting their cultural beliefs

•To comply with legislation, in particular where the death of a patient requires the involvement of a Procurator Fiscal aka. Coroner

•To minimize any risk of cross-infection to relative, health care worker or persons who may need to handle the deceased

Procedure

Often the body of the deceased is left for up to an hour as a mark of respect. The procedure then typically includes the following steps, though they can vary according to an institution's preferred practices:

•Removal of jewellery unless requested otherwise by the deceased's family. If left on it must be documented in the patient's property list.

•Wounds, including pressure sores, should be covered with a waterproof dressing. Tube insertion points should be padded with gauze and tape to avoid purging.

•The patient is laid on his/her back with arms by their side (unless religious customs demand otherwise). Eyelids are closed.

•The jaw is often supported with a pillow or cervical collar.

•Dentures should be left in place, unless inappropriate.

•The bladder is drained by applying pressure on the lower abdomen. Orifices are blocked only if leakage of body fluid is evident.

•The body is then washed and dried, the mouth cleaned and the face shaved.

•An identification bracelet is put on the ankle detailing: the name of the patient; date of birth; date and time of death; name of ward (if patient died in hospital); patient identification number.

•The body is dressed in a simple garment or wrapped in a shroud. An identification label duplicating the above information is pinned to the wrap or shroud.

•A stretcher draw sheet is placed under the body to enable removal to a trolley for transportation to the morgue. These trolleys may often be disguised to resemble laundry carts if transportation has to pass through areas where members of the public may be present.

Self-Assessment

A. Short response Questions

1. What are the classes of wastes generated in Operation theatre
2. How would you dispose wastes generated in operation theatre using color codes bin with bin liners
3. Counting is essential in accountability of used materials in operating room
4. Name three drugs used as reversal agents in operating theatre
5. What is an incident report
6. Which position would you place a patient after the following surgeries
    a) Laparotomy
    b) Tonsillectomy
    c) Transthoracic surgery

Practical task

1. Visit a skills laboratory and practice the following tasks
    a) Provide last office care to the mannequin provided
    b) Practice lifting unconscious patients from the operating table to the stretcher/from the stretcher to the PACU bed

Tools, Equipment, Supplies and Materials

Table 3 showing list of supplies and materials

Supplies	**Materials**
• Gowns	• Safety boxes
• Gloves	• Waste bins
• Masks	• Cleaning materials
• Boots	• Water
• Scrubs	• Soap/sanitizers
• Mannequin	Bed

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Learning Outcome 4: Evaluate Perioperation Theatre clinical services

Introduction to the learning outcome

This unit describes the competencies required to provide theatre clinical services. It involves Peri-Operative monitoring of patients and reporting patient's condition.

1. Patient's condition is checked & recorded according to procedure performed and Standard Operation Procedures
2. Condition of the patient is reported to appropriate personnel according to observed behavior

1.2.4 Information Sheet

Definition of terms

Parameters -a numerical or other measurable factor forming one of a set that defines a system or sets the conditions of its operation.

Content/methodology/process

Monitoring the Patient

The standard of care indicates that a registered nurse is responsible for monitoring the cardiac and respiratory status of a patient receiving local anesthetic, with or without intravenous (IV) conscious sedation, if an anesthesia provider is not present. The nurse is expected to interpret the monitoring equipment, assess the patient, and initiate interventions promptly if the patient has an untoward reaction. Policies and procedures are delineated by each facility for care of the patient receiving local anesthetic. The nurse who is monitoring the patient should not be assigned circulating duties that would distract attention from the patient.

Monitoring implies keeping track of vital functions. During extensive surgery with the patient under anesthesia, the body is subjected to physiologic stress. Bleeding, tissue trauma, potent drugs, large extravascular fluid shifts, multiple transfusions, and a surgical position that may inhibit breathing and circulation all contribute to altered physiology. These factors can induce significant cardiopulmonary dysfunction. Patients under local anesthesia are frequently monitored with electrocardiographs (ECG).

Figure 4: Diagram showing ECG segments

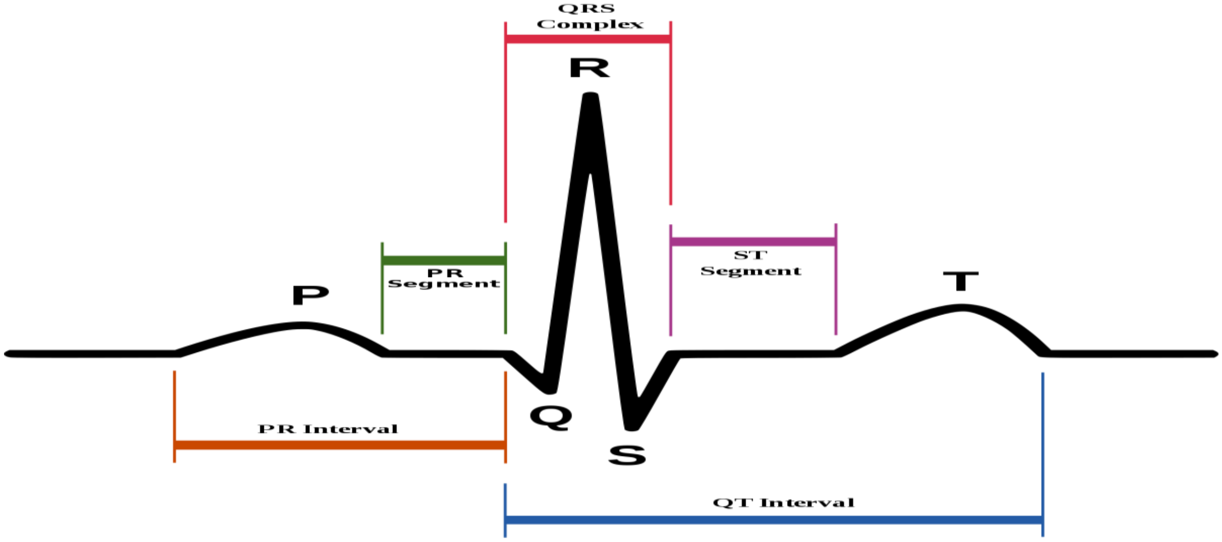

Figure 5: A patient monitor showing an ECG wave

If the patient has a known cardiac condition, such as atrial fibrillation (AF), the atrial dysrhythmia is felt by the patient and demonstrated on the ECG machine. If the patient has had atrial ablation treatments, an AF rhythm may be measured by the ECG monitor and not felt by the patient. It is important to use subjective and objective measures to evaluate the patient's condition. Evaluation of the patient's responses to surgical stressors includes observation, auscultation, and palpation. The assessment is enhanced with the use of electronic or mechanical devices that reveal physiologic trends and subtle changes and indicate responses to therapy. Most of this equipment is expensive and is incorporated into the anesthesia machine. Compact precise devices with miniaturized circuitry, better visibility, and easier maintenance are incorporated into portable models that allow for continued monitoring after the patient leaves the OR. Computers improve and expedite analysis of data. They are sophisticated data collection and management tools that assist in physiologic assessment, diagnosis, and therapy. Clinical computers vary from single-function devices to complex, multifunction, real-time systems that acquire, store, and display data; organize information; display trends; and perform calculations. Personnel who use electronic and computerized monitoring equipment must understand its use and function, be experienced in its interpretation, and be able to determine equipment malfunction easily. Instrumentation should augment, not replace, careful observation of the patient. Monitoring equipment can be inaccurate. Information from monitors should be compared with physical assessment data. Perioperative nurses involved with patient monitoring should remain current in the knowledge of physical assessment and the use of the equipment. Periodic competency testing should be performed. Written policies, procedures, and guidelines should be available for reference. The spectrum of monitoring devices is broad. It ranges from noninvasive to invasive. Noninvasive monitors do not penetrate the body or a body orifice. Conversely, invasive monitors penetrate skin or mucosa, or they enter a body cavity. Some parameters can be measured with both noninvasive and invasive methods. Perioperative nursing responsibilities may include assisting with sophisticated hemodynamic monitoring to evaluate the interrelationship of blood pressure, blood flow, vascular volumes, and physical properties of blood, heart rate, and ventricular function. Detection of early changes in hemodynamics allows prompt action to maintain cardiac function and adequate cardiac output. Monitoring facilitates rapid, accurate determination of decreased perfusion. It reflects immediate response to therapeutic measures and stress.

Physiologic Parameters Monitored Noninvasive methods can be used to monitor some cardiopulmonary and neural functions and to determine body temperature and urinary output. Both noninvasive and invasive techniques are used for monitoring hemodynamic parameters to show minute-to-minute changes in physiologic variables. Normal ranges of hemodynamic parameters are given in electrocardiogram.

Every heartbeat depends on the electrical process of polarization. Muscles in the heart wall are alternately stimulated and relaxed. An ECG is a recording of electrical forces produced by the heart and translated as waveforms .It shows changes in rhythm, rate, and conduction, such as dysrhythmias, appearance of premature beats, and block of impulses. An ECG does not provide an index of cardiac output (CO). Cardiac monitoring has become standard procedure in the OR and post anesthesia care unit (PACU). Cardiac monitoring systems generally consist of a monitor screen; a cathode ray oscilloscope, on which the ECG is continuously visualized; and a printout system, which transcribes the rhythm strip to paper to permit comparison of tracings and provide a permanent record.

Arterial Blood Pressure

Blood pressure (BP) signifies the pressure exerted against vessel walls to force blood through the circulation. Evaluation of BP during anesthesia requires consideration of blood volume, CO, and the state of the sympathetic tone of the vessels. Tissue perfusion is dependent on these factors. Arterial BP is used to assess hemodynamic and respiratory status during every surgical procedure, with very few exceptions. The BP measures contraction of the heart (systolic pressure) and relaxation of the heart between contractions (diastolic pressure). Blood is forced through the arteries between contractions. Pressure is higher during systole and lower during diastole. Many factors can cause BP to vary. It is measured in millimeters of mercury (mm Hg). Normal range is 90 to 130 mm Hg systolic and 60 to 85 mm Hg diastolic. The arm used for measurements should be opposite the one cannulated for IV fluid therapy or invasive monitoring and should be protected from contact with team members standing beside the OR bed. BP readings can be obtained using indirect or direct methods, either intermittently or continuously. Sphygmomanometer. A pneumatic cuff is wrapped around the circumference of the upper arm. When the cuff is inflated, measurements are obtained on the sphygmomanometer as the cuff deflates. Through a stethoscope placed over the brachial artery distal to the cuff, the systolic pressure is heard when blood begins to flow through the artery. The diastolic pressure is noted by a change in sound. This is an indirect noninvasive method for intermittent monitoring of BP. Doppler Ultrasound Flowmeter. With ultrasound, BP can be monitored automatically at preset intervals. The Doppler ultrasound flowmeter monitor automatically inflates the cuff, takes a reading, and deflates the cuff. It can be programmed to sound an alarm if systolic pressure reaches a preset high or low level. The readings are more accurate than with sphygmomanometer pressure cuff monitoring because the ultrasonic transducer amplifies blood flow sounds. Automated noninvasive BP monitors function well in a noisy environment. However, they do not provide continuous BP measurement or detect extremely low pressures. Frequent inflations of the cuff can bruise the skin, especially of geriatric patients. Infrared Beams. Noninvasive infrared beams (Finapres technique) may be used to indirectly monitor BP. A small cuff fits on a finger. The beams from the cuff shine through tissue. They continuously measure arterial pressure from one heartbeat to the next. Direct Arterial Pressure. From an invasive modality, beatto-beat direct pressures are obtained through an artery, usually the radial or femoral, via an indwelling catheter inserted percutaneously. A very slow drip of slightly heparinized sterile saline solution keeps the catheter open. The fluid-filled tubing from the catheter is connected to a mechanical electric transducer. The transducer is attached to an amplifier.

Blood Gases, Oxygenation, and pH Monitoring of tissue perfusion is indispensable in the evaluation of pulmonary gas exchange and acid-base balance. Measurements are considered in relation to other parameters, such as vital signs, venous pressure, and left atrial pressure. Oxygen and carbon dioxide in blood exert their own partial pressures (P). Measurements are expressed in millimeters of mercury or torr. They may be differentiated as arterial (Pa) or venous (Pv). The gas being measured is identified. The partial pressure of oxygen is expressed as Po2, or specifically in arterial blood as Pao2 and in venous blood as Pvo2. Oxygen saturation (So2) in arterial blood (Sao2) is expressed in percentages. The partial pressure of carbon dioxide is specified as Paco2 or Pvco2. Monitoring techniques can be either noninvasive or invasive. Pulse Oximetry. A pulse oximeter measures arterial oxyhemoglobin saturation (Sao2). It provides a reading within seconds by measuring the optical density of light passing through tissues. A sensor probe is clipped on each side of a pulsating vascular bed. Fingers, toes, earlobes, and the bridge of the nose are suitable sites. The patient's skin should be clean and dry. An area with fingernail polish should be avoided or the polish removed. Skin integrity under the sensor must be intact.

Central Venous Pressure Because it accurately measures right atrial BP, which in turn images right ventricular BP, central venous pressure (CVP) monitoring assesses function of the heart's right side. It measures the pressure under which blood returns to the right atrium and reflects pressure in the venous system as blood returns to the heart. In other words, CVP represents the amount of venous return and filling pressure of the right ventricle. This information helps determine the patient's circulatory status. CVP monitoring also aids in the evaluation of blood volume and the relationship between circulating blood volume and the pumping action of the heart (i.e., adequacy of volume presented to the heart for pumping). Therefore CVP monitoring is a useful guide in blood or fluid administration to avoid circulatory overload in patients with limited cardiopulmonary reserve. Too great or too rapid replacement can cause pulmonary edema. Generally, a low CVP indicates that additional fluid can be given safely. CVP monitoring may be used during shock or hypotension to judge the adequacy of blood replacement. However, CVP is not a measure of blood volume per se or of CO. Indications for CVP monitoring include major surgical procedures in patients with preexisting cardiovascular disease, in surgical procedures in which large-volume shifts are anticipated (e.g., open heart surgery), in critically ill patients (e.g., massive trauma, malignant hyperthermia), in surgical procedures in which venous air emboli are a risk (e.g., craniotomy in a sitting position), and in rapid administration of blood or fluid. Although CVP monitoring provides valuable data for assessment of the adequacy of vascular volume, it only indirectly reflects the function of the left side of the heart. There is no direct relationship between right and left ventricular filling pressures. Because of the dispensability (compliancy) of the pulmonary blood vessels, the lungs can accept a marked increase in blood flow before significant congestion appears. Backup of blood caused by impaired function of the left ventricle and a subsequent increase in pulmonary vascular resistance (PVR) may occur before this increased pressure affects the right side of the heart, as exhibited by CVP values. Thus, CVP does not correlate with left-sided heart performance in patients with left ventricular dysfunction or pulmonary congestion. Central Venous Cannulation. CVP may be monitored with a single-lumen or multilumen radiopaque catheter. A double-lumen or triple-lumen Hickman or Broviac catheter is used most commonly. The right atrial lumen of a Swan-Ganz pulmonary artery catheter also can be used to obtain CVP readings. The lumens are labeled and color-coded on multilumen catheters. The catheter is inserted, preferably percutaneously through a subclavian vein. A brachial, external or internal jugular, or femoral vein may be used or, via cutdown, the antecubital vein. If patients are awake, they are asked to bear down (Valsalva's maneuver) as the vein is punctured. This increases intrathoracic pressure and counteracts negative pressure from the vein, thus reducing the possibility of air embolism. The catheter is threaded through the vein and advanced into the superior vena cava or right atrium. This may be done by fluoroscopy, or a chest x-ray may be taken to verify accurate placement of the catheter tip. The catheter may be attached to a transducer and monitor. Pressure readings are expressed in millimeters of mercury. The catheter can be attached to a fluid-filled manometer that measures pressure in centimeters of water (cm H_2O). To set up this line, the IV solution bag is connected to the tubing, the manometer is inserted into the line by attaching it to the stopcock between the IV tubing and the extension tubing, air is expelled from the line, the line is clamped, and the manometer is secured upright to an IV pole. The hub of the catheter lumen is connected to the stopcock. Connections to the three-way stopcock should be taped to prevent inadvertent disconnection and air leaks. Cyclic variations in intracaval venous pressure occur; pressure becomes negative during atrial filling and respiratory inspiration. Sucking of air into the system during negative venous pressure can result in an air embolus. Baseline measurement is taken as soon as the catheter is in place and attached to the monitor. This is also a presumptive check for proper placement of the cannula tip. Because expansion of the lungs increases intrathoracic pressure and deflation decreases it, fluid in the manometer should fluctuate with each breath. To obtain a CVP reading on the manometer, adjust the scale to zero level with the patient's right atrium. The lumen of the stopcock to the catheter is shut off, which allows the IV fluid to run into the manometer to the desired level. The infusion is shut off, and the catheter is opened. After the reading is obtained, the infusion lumen to the catheter is opened to keep the line open. Electronic transducer systems provide continuous monitoring of venous pressure. Continuous monitoring supplies good measurement of the right side of the heart and portrays the trend of heart function that is more valuable than are isolated readings obtained with a manometer. CVP values may vary somewhat, but normal readings range from 2 to 8 mm Hg, or 3 to 10 cm H_2O.

Cardiac Output

Cardiac output is measured using the thermodilution technique to determine liters of blood pumped per minute by the left ventricle into the aorta. Normal resting value is 4 to 8 L/min. A known amount of fluid at a known temperature is injected into a lumen of an arterial catheter, and a temperature gradient at a point downstream is measured via a second lumen. Iced-cold or room temperature physiologic saline solution or 5% dextrose in water (10 mL) generally is used. Blood flow supplies the thermal dilution; for example, saline solution mixes with blood in the superior vena cava or right atrium, depending on the catheter location, which reduces the temperature of blood in the heart. The cooled blood flows past a transistorized intravascular thermistor in the thermodilution catheter that detects changes in blood temperatures that are then used to compute CO.

When the solution is injected via the proximal right atrial lumen of a pulmonary artery Swan-Ganz catheter, a digital display of CO is seen within 4 to 5 seconds. CO reflects the mechanical activity of the heart and represents total blood flow to all tissues and vascular shunts. It depends on the heart rate, the contractile strength of the heart muscle (myocardial contractility), the peripheral resistance of vessels, and venous return. Inotropic agents such as digitalis or epinephrine increase contractility and CO, except in patients with loss of functioning ventricular muscle (e.g., after myocardial infarction or an aneurysm of the left ventricle

In compromised hearts, this pressure is higher. A reduced CO results in decreased perfusion of the capillary circulation. During hemorrhage, when circulating blood volume is reduced, the resulting diminished venous return and preload lead to a lowered CO. Atrial fibrillation also can modify the filling of the ventricles. Venous dilation contributes to pooling of blood, with subsequent decreased venous return to the heart. Low CO states result from reduced preload, as in hypovolemia, venous dilation, or cardiac tamponade; reduced contractility, as from anesthetic drugs, ischemia, infarction, or cardiac decompensation; dysrhythmias; or increased afterload, as in hypertension, pulmonary emboli, or an elevated or diminished heart rate.

Total Blood Volume

Blood volume is useful in determining the total amount of blood replacement necessary. An accurate method of total blood volume (TBV) measurement involves measuring plasma and red blood cell volumes separately and then adding the results together. For measurement of red blood cell volume, cells are tagged with detectable, nontoxic, radioactive chromium, subsequently injected intravenously, and counted after an appropriate mixing time. Or radioactive iodinated human serum albumin, in standard-dose packages, can be injected, mixed, and counted. Counting may be done rapidly using an electronic device. This technique may be used in place of estimation of blood loss.

Respiratory Tidal Volume

The respiratory tidal volume (VT), the volume of air moved with each respiration and expiration, may be measured with a respirometer placed on the expiratory limb of the anesthesia machine or mechanical ventilator. Alarms may be incorporated to signal disconnection, failure to cycle, and excessive pressure.

Body Temperature

The body continuously produces heat through metabolic activities and loses heat through convection, evaporation, conduction, and radiation. The production of heat causes increased oxygen consumption by the body’s cells. When the rate of heat production is equal to the rate of loss, a heat balance of constant core body temperature is maintained. Core temperature is that of the interior of the body as opposed to the body surface temperature. Normal core temperature ranges from 98° F to 100° F (36.8° C-37.7° C). Under anesthesia, the average adult loses 0.9° F to 2.7° F (0.5° C-1.5° C) of body temperature; the greatest loss occurs during the first hour, through convection from exposure to the environment, through evaporation via respiration, through conduction from contact with cool surfaces, and through radiation from tissues. Intraoperative hypothermia, or core temperature less than 96° F (36° C), is a common complication of surgery under general anesthesia, especially in pediatric and geriatric patients. Some anesthetic agents inhibit heat production: halogenated agents cause vasodilation that contributes to surface cooling and muscle relaxants prevent shivering, which is a thermoregulatory protective reflex. Other factors can change core temperature.

Urinary Output

Urinary output can be measured with an indwelling Foley catheter attached to a calibrated collection bag. The sterile disposable collection system must be below the level of the bladder to prevent distention and allow flow without reflux. Output is valuable in assessing effective blood volume and fluid administration, except when a diuretic is given. Volume, electrolytes, osmolarity, and pH are important. Any change in urine color can signal a change in the patient's condition. Brown urine can mean the presence of hemoglobin or myoglobin caused by rhabdomyolysis in a hypermetabolic crisis, such as malignant hyperthermia. Frank blood or clots may indicate an injury to the kidney, ureter, or bladder from iatrogenic cause or other trauma. IV dye, such as methylene

Table 4: Showing the patient monitoring parameters with normal ranges

Parameter	Abbreviation	Normal Range For Adults
Arterial oxygen content	CaO_2	17-20ml/dl
Blood pressure	BP	Systolic 90-130mmHg Diastolic 60-85mmHg
Cardiac output	CO	4-8l/min
Central venous pressure	CVP	2-8mmHg
Glomerular filtration rate	GFR	80-120mls/min
Heart rate	HR	60-100 beats per minute
Intracranial pressure	ICP	0-15mmHg
Oxygen saturation in arterial blood	SaO_2	95-97%
Partial pressure of Oxygen in arterial blood	PaO_2	80-100mmHg
Stroke volume	SV	60-130Mml/beat

Standards and Methods for Documentation of Patient Care

The standards for patient care documentation are established by the American Nurses Association (ANA) and TJC. The standard of care requires that patient care documentation reflect the application of the nursing process (assessment, nursing diagnosis, outcome identification, planning, intervention, and evaluation) during the entire length of stay, according to ANA and TJC.

The use of the Perioperative Nursing Data Set (PNDS) is the method of choice for perioperative patient care documentation. The PNDS provides a standardized universal language for patient care documentation and is used by many surgical computer information system manufacturers. Charting Modalities Many ways of recording patient care information have been used over the years. Changes in technology have created more methods of recording patient care; below is a list of examples.

Narrative charting

Writing about significant events using third-person commentary, quotes, and standard abbreviations. Entries are sequential, timed, dated, and signed.

Block charting

Short commentary on activity that resembles narrative charting covering a longer period of several hours or days. Entries are sequential, timed, dated, and signed. Some facilities use a checklist format.

Focus charting

Specific documentation directed at a designated aspect of the patient's needs, status, or health considerations.

Subjective-objective charting (subjective-objective assessment plan [SOAP]). Multidisciplinary approach to documenting care according to cues given by a patient with a specific set of signs and symptoms. This approach uses direct quotes and assessment data. Problem-oriented charting (problem-oriented medical record [POMR]). Approach using a problem list as the working element from which care is planned. Working from the list, patient priorities are investigated, diagnosed, treated, minimized, solved, or remain ongoing. As problems are solved, they are stricken from the list.

Computer-generated charting. Use of standardized care plans formulated in the computer and modified for the individual patient. The computer time and date stamps the plan as it is printed for the hard copy record. This form of charting requires the caregiver's signature.

Computer information systems, check-off forms, and flow sheets. Commonly used as shortcuts for record keeping. Unfortunately, it is easy to rely on the standardized data on these preprinted records and inadvertently omit potentially important individualized information. Most computerized patient records have secondary documentation fields to complete for individualized data capture.

Computerized Documentation- Many facilities have been using computers for patient admitting, billing, scheduling, and human resource information for several decades. Within the past few years, patient data have been recorded and stored electronically at the patient care unit level. Referred to as electronic health records or electronic medical records, the documentation is transmitted electronically to multiple sites including the OR, surgeon's offices, and patient care areas. Only select personnel are permitted to access this information and must log on using employee identification and passwords to enter the computer system.3 Passwords are changed at routine intervals. Retrievable data recorded and accumulated in these electronic files include patient care information, laboratory results, surgical reports, admission and discharge summaries, and many highly sensitive details about a patient's financial status. Many larger multihospital systems, such as the Cleveland Clinic, have utilized the Internet to permit multiple record access points for office based physicians, surgeons, and the patient. All authorized users have access to reports and health data as soon as they are entered into the system. Nurses charged with the responsibility of accessing and contributing to computerized patient data should be aware that security and confidentiality must be protected. Failing to maintain the secrecy of passwords and failing to log off after use are common problems identified with unauthorized access. Some systems have a built-in log-off feature if the workstation is left idle for a prolonged period. If this happens, the user has to reenter the system by logging back on. Perioperative Documentation Specific care given in the perioperative environment should be documented on the patient's chart. Most facilities use a preprinted form with a standardized plan of care. Space is provided to add individualized patient needs and to document additional interventions. Data included in the record come from several patient care areas. A comprehensive checklist is included with the chart to assist the circulating nurse to determine whether all of the data

Patient Hand-Off or Handover

- A hand-off is defined as a linear transmission of information from one person to another.
- The Joint Commission recommends a standardized hand-off in their National Patient Safety Goals. It is called the Targeted Solutions Tool for Hand-off Communications.
- AORN also has recommendations for transferring patient care information.
- Standardized protocols are recommended because the hand-off or handover of patients to another health care provider is reported as a high-risk time associated with sentinel events.
- Every facility should establish a format for standardized hand-off protocols and policies. Standardized protocols provide critical patient information necessary for safe patient transfer and reduction in communication breakdown.
- Many standardized models exist. An example of a standardized documentation format is SBAR (Situation, Background, Assessment, and Recommendation).
- Another popular documentation format is SHARED (Situation, History, Assessment, Request, Evaluate, and Document).
- Hand-off reporting improved when health care facilities established a format of standardized hand-off protocols. These protocols are written, verbal, and electronic methods of documentation and communication.
- Hand-off communication improved when personnel were educated about the standardized protocols and policies. Positive education methods include: role playing, competency training, case studies, anticipated patient needs, and simulation.
- Ways to improve hand-off include: reduce noise and distractions, take notes, use a checklist, improve listening skills, use standard vocabulary, provide accurate documentation, document person taking hand-off report, ask questions, and read back information to verify. CONS
- Communication breakdown has been documented as the root cause of many sentinel events.
- Problems and errors related to poor hand-off include: room noise, rushing, errors on record, distractions, change of staff, lack of information, personnel physiologic problems, relationship barriers, lack of experience, and number of personnel involved in the hand-off process.

• More research is needed to identify an error-free transfer of information to improve patient outcomes. The current standardization protocols still have breakdowns in communication putting patients at risk for injury.

Self-assessment

1. Define patient hand over?
2. Post recovery, what aspects of urine monitoring can be used to monitor status of a patient
3. What is the normal temperature range
4. How does general anesthesia affect patient temperature
5. What is cardiac output
6. What is the difference between systolic and diastolic pressure?
7. When monitoring status of a patient, an electrocardiogram indicates the functioning of a patient, what are the normal ECG waves on an electrocardiogram?
8. What type of documentation can be done in operating theatre?

Practical tasks

1. Visit a skills laboratory session and demonstrate how to take blood pressure using sphygmomanometer.
2. Draw an ECG wave to illustrate a normal heart activity

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